Communicating with Models and Data

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October 13, 2026
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Abstract

Many real-world decisions rely on external data collection and modeling: consultants provide clients scenarios, fund managers present models of asset returns, and AI chatbots offer recommendations based on internal data. I study how model simplification in these settings distorts not just how data are interpreted but also which data are collected. A sender collects data and estimates a model that helps a receiver make a choice. He would like the receiver to be optimistic about that choice. He is either "accurate," and correctly estimates the tradeoffs between choices and outcomes, or "simple." A simple sender's model artificially restricts the curvature of the tradeoff between choices and outcomes: a "naive" simple model neglects this curvature, while a "cautious" simple model exaggerates it. I show that estimating simple models creates strategic incentives to collect data poorly suited to the receiver's decision. Senders who estimate naive models sample data about an excessively large range of choices, while those who estimate cautious models sample locally around a single extreme choice. With multiple variables, senders sample data that maximize collinearity across variables. These distortions make model simplicity strategically valuable: accurate senders benefit when receivers perceive their models as naive or cautious. In equilibrium, accurate senders surround themselves with simple senders who polarize into two groups: those who estimate maximally naive models and those who estimate maximally cautious ones. In the limit, these correspond to linear regressions and "worst-case scenario" models, respectively.

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1 Introduction

Consider a policymaker tasked with designing a carbon tax. She understands the nonlinear tradeoff between benefits and costs: low taxes do not curb emissions, while high taxes generate costs. The optimal tax lies between these extremes, but she does not know where. So, she hires a consultant, who collects data and estimates a *model* mapping taxes to welfare. But while his model is valuable, it presents three frictions. First, it may be too simple. Suppose the consultant estimates a linear regression of welfare on taxes. This implies the optimal tax is either very low or high, *neglecting* the tradeoff between benefits and costs. Or, he might estimate a "worst-case scenario" that *exaggerates* the tradeoff by always assuming high costs. In either case, the policymaker believes the model is simplistic, and thus tries to infer the underlying data. This presents a second friction: she cannot observe that data, which may be proprietary or reflect expertise, but which also may be more informative about some taxes than others. Third, the consultant may have financial or reputational incentives to make her optimistic about how much she can improve welfare.

This example reflects a broader phenomenon: decisionmakers rely on others to model the world for them. CEOs hire consultants to guide firm restructuring, investors hire fund managers to model returns, voters use political narratives to understand laws, and, increasingly, any of these decisionmakers may utilize AI chatbots to inform choices. A growing literature documents that these models are often *simple* by design. Financial professionals' "rules of thumb" (Shiller, 2017; Bastianello et al., 2024) and managers' simplified models of price-setting leave profits on the table (DellaVigna and Gentzkow, 2019; Han et al., 2026). Moreover, their data may be opaque and non-generalizable: experts choose which data to collect (Di Tillio et al., 2021), and AI chatbots may lean on unobservable training data (Jin et al., 2024). Yet these simple models and narratives drive real choices: they may generate optimistic hiring (Flynn and Sastry, 2025), political polarization (Levy et al., 2022; Herrera and Trombetta, 2026), or media engagement (Eliaz and Spiegler, 2026). How do *model simplicity and strategic data collection* shape what decisionmakers learn?

This paper studies a theoretical setting where a receiver (she) hires a sender (he) to help her make a choice. The sender gathers data and estimates a *model* mapping choices to outcomes. He would like to make the receiver optimistic about the highest outcome she can achieve. The sender has a private type determining whether he is "accurate" or "simple." An accurate sender correctly estimates the functional form between choices and outcomes. A simple sender estimates a model that artificially *fixes* the curvature of the functional form: he either *neglects* tradeoffs and assumes low curvature (as in a linear regression) or *exaggerates* them and assumes high curvature (as in a worst-case scenario). I show that a simple sender can strategically sample data to make the receiver optimistic, since she entertains other models that could have produced those data. If he neglects tradeoffs, he samples data about an excessively wide range of choices; if he exaggerates them, he samples locally around an extreme choice. An accurate sender benefits when the receiver suspects his model might be simple, since she entertains models more optimistic than the truth. In equilibrium, accurate senders thus surround themselves with a polarized group of simple senders who fix curvature near either the lowest or the highest values the receiver finds plausible.

Model Overview A receiver — such as a policymaker, CEO, investor, or voter — must choose an action x to maximize an outcome y — such as profits or the value of a public good. She believes that the true functional form mapping x to y is given by a set of data-generating-processes (DGPs) that capture a tradeoff between a linear benefit and a convex cost. The magnitude of the convex cost is given by a parameter $\gamma \in [0, \overline{\gamma}]$, which represents the *curvature* of tradeoffs.

The receiver hires a sender to acquire information for her. First, the sender chooses a distribution of x values to sample from, which the receiver does not observe. He then observes the corresponding y values, producing data (y, x), and fits a model to the data. The sender has a private type that determines the models he can fit. He is either "accurate" or "simple." An accurate sender considers the same models as the receiver. A simple sender considers similar models, but they are *misspecified* and treat tradeoffs in a simplified way. Formally, he assumes the curvature parameter γ has a *fixed value*. If he assumes γ is always lower than the receiver expects, we say he estimates a "naive" model, which *neglects* tradeoffs the receiver considers. For example, he may assume $\gamma = 0$, in which case his model is a linear regression of y on x. If he assumes γ is higher than the receiver expects, we say he estimates a "cautious" model, which *exaggerates* tradeoffs between choices and outcomes. For example, he may assume $\gamma = \overline{\gamma}$, i.e. that the marginal costs of choices are as high as possible, corresponding to a "worst-case scenario" model.

The receiver is Bayesian: she observes the sender's model, infers whether he is accurate or simple and how he collected data, and updates over the set of models she considers. The sender's objective is to make the receiver optimistic, i.e. to make her believe the maximum value of y she can expect to achieve is as *high as possible* — because he would like to maximize her engagement with his services or receives a cut of her expected profits. He knows that the receiver might entertain additional models that will affect her choice, which shapes his initial data sampling decision.

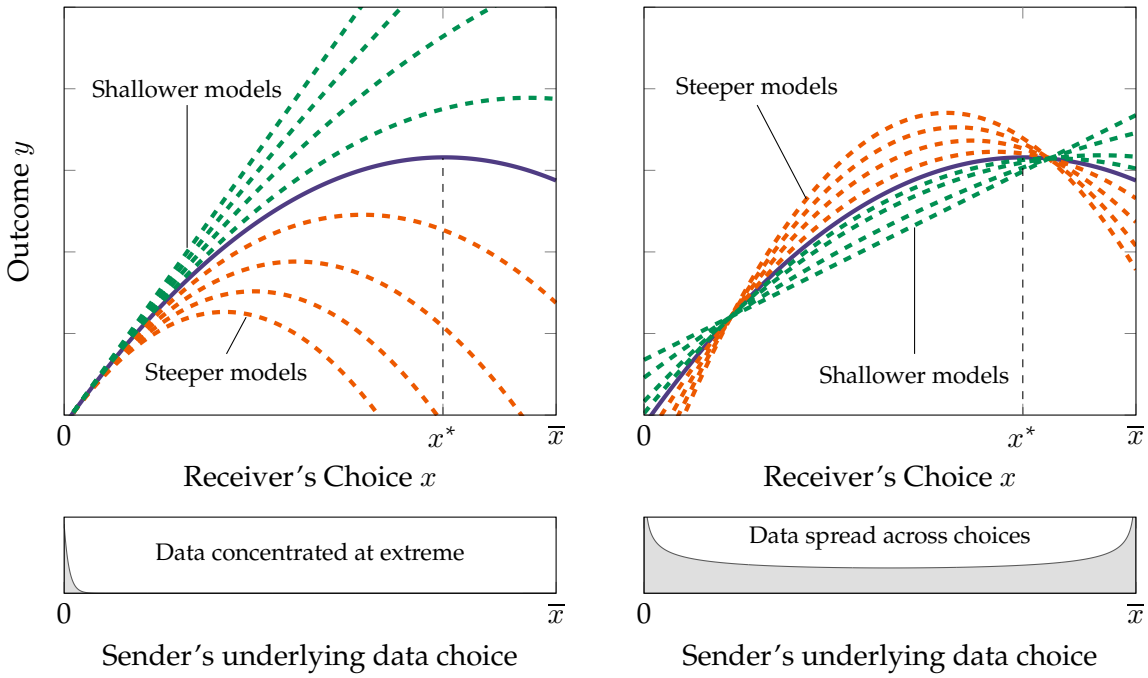
Receiver Learning and Sender Sampling The first result is that receiver learning is shaped by the interaction of the sender's model simplicity with strategic data collection. The sender provides a model, shown in purple in the figure below. It is possible that the model came from the accurate sender, and is an accurate depiction of the true DGP. However, because the model may be simple, the receiver may entertain alternative models. These plausible models have *steeper* or *shallower curvature* than the original model.

I highlight two particular forms of data collection. The first is when a simple sender positions himself as a "specialist" by estimating models "locally" on a small neighborhood of x values. In this case, the receiver's set of plausible models is tangent to the sender's model at a single point. This is illustrated in panel (a) of Figure 1. At x^* — where the simple sender's model is maximized — steeper models lie *below* the sender's model and shallower models *above*.

As the set of choices over which the sender samples widens, the receiver's additional models shift up or down relative to the sender's model, depending on their curvature. The second form of data collection is when a simple sender estimates models across the largest possible range of x values. I refer to this as "generalist" sampling. In this case, the models the receiver entertains

are *secants* through the sender's original model. Steeper models lie *above* the original model and shallower models lie *below*. This is shown in panel (b) of Figure 1.

Figure 1: Models receiver entertains when the sender provides simple model



(a) Plausible models under "specialist" data

(b) Plausible models under "generalist" data

This leads to a second set of results: when the sender presents a *naive* model, he benefits from positioning himself as a generalist. Formally, I show that this sender samples data centered in the middle of the receiver's choice space with an excessively high variance, which delivers the picture in panel (b) of Figure 1. The intuition is that if the sender presents a naive model, he is neglecting tradeoffs the receiver finds important. This means that the additional models the receiver entertains will be *steeper* than the original model. Steeper models lie above the sender's original model when the sender positions himself as a generalist. Thus, the sender convinces the receiver she can achieve an even higher maximal expected utility than his original model.

The opposite happens when the sender presents a *cautious model*, where he benefits the most from positioning himself as a specialist. In this case, any additional receiver models are shallower than the original model, which lie above it when he samples as a specialist. Moreover, I show that this sender samples in the smallest possible neighborhood of either the receiver's maximal or minimal choice. Panel (a) of Figure 1 shows a specialist sender sampling close to $x = 0$. Sampling thus has a "bang-bang" property: the sender either samples data precisely about a fringe receiver choice or across the widest possible range of choices, rather than an intermediate amount.

Third, I extend the baseline model to account for multidimensional choices and show that senders collect data that maximizes collinearity between variables. If a sender neglects a (negative)

interaction effect between two variables, he samples data so they are as correlated as possible. If he exaggerates interactions, he samples so that those variables are as anticorrelated as possible. The intuition is that if the sender's model neglects interactions, the sample mean observed in his data is in part driven by a negative interaction of two variables that are highly correlated. If the receiver were to unwind this correlation, the sample mean would be higher than what she initially observed, meaning her maximal expected utility would also be higher. The opposite logic holds when the sender's model exaggerates interactions. While the paper abstracts from finite samples, this means that senders who estimate simple models collect datasets that are heavily unbalanced, generating imprecise model parameter estimates.

Fourth, I show that a receiver's choices are more accurate under an intermediate degree of sampling, rather than specialist or generalist sampling. Thus, conditional on model simplification, simple senders do not even sample data in a way that is second-best for the receiver, and include either precise data on choices at the fringe of the receiver's choice space (cautious specialists) or thick-tailed information from all across the choice space (naive generalists). This low quality of data emerges precisely as a consequence of the sender's model simplification.

The baseline model assumes the sender always estimates a misspecified model with *fixed* curvature. However, I also show that the results generalize if the sender can estimate models with a range of curvatures that are still a subset of the receiver's. Moreover, while most of the paper assumes the receiver's payoff y is quadratic in x, I also show that the results are robust to the receiver's payoffs being given by a linear benefit less a symmetric, convex loss around a point. Although the model without symmetry can prove to be very intractable, I provide a regularity assumption that delivers an analogue of my result: a sender who estimates naive models gains more from being a generalist than a specialist, and one who estimates cautious models gains more from being a specialist than a generalist.

Gains from Simplification and Market for Models Fifth, I show that a sender who provides an accurate model gains from being mistaken for a sender who supplies simple models. Recall that one sender always provides an accurate model of choices, and the other provides a simple model. I show that the accurate sender gains from the existence of the simple one. The reason is that, conditional on seeing a model, the receiver believes it is either an accurate model or a simple model. In the latter case, the receiver considers models that deliver higher expected utility than the original, due to the combination of strategic sampling with tradeoff neglect/exaggeration. If a sender who provides an accurate model can plausibly masquerade as one who provides a simple model, he thus makes the receiver more optimistic than he would initially; she considers both his original (true) model and ones that deliver higher expected utility.

Using these insights, I then ask what the *market* for senders may look like for a receiver. This delivers a sixth and final result: whenever a receiver contracts the services of a sender, she will receive one of:

1. an accurate model mapping choices to outcomes;

2. a highly naive model, calibrated to a fixed value of γ close to 0;

3. or a highly cautious model, calibrated to a fixed value of γ close to $\overline{\gamma}$.

To obtain this result, I endogenously allow the accurate sender to surround himself with simple senders. In particular, he can choose the *degree* of simplicity simple senders possess (what models they estimate) and their *distribution* in the population (how many of these senders there are). For example, a consultant might choose to hire other consultants at his organization who think about tradeoffs in simpler ways; a politician may be able to propagate simple narratives in the media; or an AI chatbot may deploy simpler models alongside complex ones.

I show that, in equilibrium, the accurate sender will only choose to surround himself with simple senders who never estimate accurate models, and only ever estimate models *pre-calibrated* to a fixed value of γ. Formally, for each value of γ, the accurate sender hires a simple sender who can *only* model tradeoffs with that value of curvature. That simple sender then strategically samples data (as a generalist or specialist) so that additional receiver models always lie above the original model.

Then, I show that the *distribution* of these simple sender types is associated with *polarization* in how they consider tradeoffs. I show that the accurate sender always surrounds himself with few (or no) simple senders who model tradeoffs at intermediate values of γ. Instead, he mostly hires senders who estimate models that are (calibrated to be) very naive (with fixed γ close to 0) cautious (with fixed γ close to $\overline{\gamma}$). These senders not only provide pre-calibrated models, but they tend to be either excessively naive or cautious. In limiting cases, simple senders either *neglect all tradeoffs* and estimate *linear regression models*, or *exaggerate all tradeoffs* and estimate a worst-case model, i.e. those that assume the curvature of tradeoffs is fixed at its maximal value $\overline{\gamma}$.

Overview of Paper Section 2 lays out the basic model. Section 3 characterizes learning, presents the results on data sampling using two leading sender types for intuition, and then provides insights with multivariate choices. Section 4 extends the model to investigate the value of learning and forecast error. Section 5 studies gains from oversimplifying models and the design of the market for senders. Section 6 concludes and discusses implications. The remainder of this section reviews the relevant literature.

Literature This paper broadly adds to our understanding of how we learn from externally-provided models and data, and how incentives to make us optimistic distort the design of information through strategic data collection and model simplification. Its key contribution is to characterize what *sorts* of data model senders may strategically sample, how incentives to strategically sample data are informed by the simplicity of their models' curvature, and how the interaction of both can generate gains from systematically pursuing highly simplistic data-generating-processes.

Many of the examples in the paper concern *delegation* to experts. There is a long line of literature — particularly in organizations and finance — studying this process, including mistrust

in experts (Cheng and Hsiaw, 2022), uncertainty about expertise (Kishishita, 2020), and its interaction with persuasion (Kolotilin and Zapechelnyuk, 2025). My contribution to this literature is to study how delegation and communication operate when an expert's recommendation comes in the form of a *model* organizing endogenously collected *data*. This conceptually contrasts my paper with work like Deimen and Szalay (2019), who also study delegation to experts, but are concerned with the information that can be gleaned with experts' reports when their preferences may be misaligned with decisionmakers'. The most closely related paper in the finance literature is Li and Wang (2018), who study delegation when a portfolio manager provides a model of returns to a client.

While there are many papers that study learning from models, much of this literature takes as given the past history of data that a model is fit on — whether that data is exogenous or endogenously generated by a principal's or agent's choices (Eliaz and Spiegler, 2020; Schwartzstein and Sunderam, 2021; Levy et al., 2022). My contribution is to study a sender who first chooses a data collection rule and then provides the best-fitting model to that data. Thus, although data collection is endogenous and unobserved by the receiver, the sender's hands are still tied by his initial data collection. This assumption contrasts with other papers where data is endogenously generated either through follow-on evidence generation (Dai et al., 2026), selective disclosure (Felgenhauer and Schulte, 2014), nonrandom sampling (Di Tillio et al., 2021), or sequential data acquisition with endogenous stopping (Henry and Ottaviani, 2019). These papers show that the ability of a sender to persuade a receiver often depends on the sorts of signals about data a sender can transmit. In my setting, I show that a sender's desire to make the receiver optimistic coupled with model simplicity creates a tractable characterization of data sampling: the sender either locally samples data in the smallest possible neighborhood of an extreme receiver choice, or as distantly as possible across a wide range of receiver choices.

The use of strategic *sampling* interacts with a sender's model simplicity to make the receiver believe her expected utility is *higher* than under the model provided by the sender. This channel through which a receiver can be made more optimistic is thus related to model simplicity and data sampling, as opposed to departures from a status quo or the inclusion of confounding variables, such as in Eliaz and Spiegler (2020). This also distinguishes the paper from work where a receiver selects between different models with different sample sizes and more or less complex models (Montiel Olea et al., 2022).

My paper shows that gains from model simplicity or strategic sampling can arise without interpreting the role of noise in driving data-generating processes. This finding contrasts with papers that investigate gains from model-based persuasion that involve trading off model fit and belief manipulability — such as Schwartzstein and Sunderam (2021), Aina (2023), Barron and Fries (2024), and Aina and Schneider (2025). The present paper does, however, show that senders who make receivers more optimistic often create greater *forecast* error for a receiver, suggesting that receiver optimism may come at the cost of uncertainty over models.

The particular angle the present paper takes is closely related to omitted variable bias — in that

a sender cannot (or refuses to) fully model a tradeoff in the way a receiver would like. Thus, this paper is related to an older literature on omitted variable bias in econometrics (see Theil, 1957; Altonji et al., 2005; Oster, 2019, and references therein). It also connects the paper to theoretical work like Levy et al. (2022), Gleyze and Pernoud (2022), and Eliaz et al. (2025) — who show that model misspecification may interact with competitive pressures or uncertainty about data estimation to generate a range of outcomes from improved communication and engagement to political cycles and the prevalence of false models. More broadly, this paper studies a model where a receiver entertains a well-specified model and a sender estimates a misspecified model. It thus contributes to a deeper theoretical literature on Berk-Nash equilibrium and learning with misspecified beliefs (Alonso and Câmara, 2016; Esponda and Pouzo, 2016; Spiegler, 2016; Heidhues et al., 2018; Mailath and Samuelson, 2020; Bohren and Hauser, 2021; Fudenberg et al., 2021; De Clippel and Zhang, 2022; Levy et al., 2022; Bohren and Hauser, 2025).

A key finding of this paper is that senders may generally benefit from providing *simpler* models to receivers. The idea that senders can benefit from simple models is also captured by Perez-Richet and Prady (2012), Bloedel and Segal (2018), Lipnowski et al. (2020), Eliaz et al. (2021), and Burkovskaya and Starkov (2026). While many of these papers are driven by behavioral assumptions like bounded rationality or rational inattention, the findings in this paper are related to sampling and simplicity: given a sender's model, a receiver will entertain additional models. The set of these additional models is determined by the *simplicity* of the sender's model, and the expected utility they generate for the receiver increases or decreases with the spread of the sender's data sampling. In particular, the present paper shows that a sender informed of the true model of the world can benefit by making the receiver think his model was estimated by a sender who had an artificially simple model *and* strategically sampled. The most closely related paper is Gleyze and Pernoud (2022), who show that an informed sender can often gain by delegating model provision to a (potentially misspecified) sender.

Finally, because of its reliance on models as information structures, the present paper also adds to a longer literature on (Bayesian) persuasion (Kamenica and Gentzkow, 2011), as well as persuasion and competition (Gentzkow and Kamenica, 2017; Wang, 2013). In particular, receivers in the present paper are uncertain as to the source of data or the broader class of models a sender entertained. This connects the paper to models of persuasion and cheap talk with *uncertainty* about information structures (Huang, 2024; Min, 2026) and ambiguity aversion (Colo, 2024).

2 Model Setup

I begin by laying out a game between a single receiver R (she) and a single sender S (he). R must make a choice to maximize an outcome, and enlists the services of a sender S. S collects data and provides R a (potentially misspecified) model describing that data. R sees S's model, updates, and then makes the choice that maximizes the expected outcome. To help map the model to the real world, I use the running example from the introduction where R is a policymaker tasked with

designing a carbon tax, and S is a consultant who models the effects of different taxes.

2.1 Receiver and Models

R faces a set of choices $X = [0, \overline{x}] \subseteq \mathbb{R}$ for $\overline{x} > 0$. Each of these choices maps to an outcome $y \in \mathbb{R}$. Ex-ante, R does not know the relationship between y and x. However, she believes that the value of y given x can be described by a data-generating process (DGP) $y = f_R(x)$ for a function $f_R : X \to \mathbb{R}$. We refer to f_R as a **model**. We denote F_R as the set of *models* f_R that R entertains.

I assume that for each $f_R \in F_R$, we have

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma x^2,$$

where $\alpha_0 \sim p_0^R$ on $\mathbb{R}$, $\alpha_1 \sim p_1^R$ on $\mathbb{R}$, and $\gamma \sim p_\gamma^R$ on $[0, \overline{\gamma}] := \Gamma$ for continuous priors p_0^R, p_1^R, p_γ^R. Thus, each of R's models is concave, and can be represented as an affine function less a convex, quadratic loss.[1] The magnitude (or curvature) of this quadratic loss is governed by the parameter γ. Her prior over each of the three components is independent.

The receiver's goal is to maximize the expected value of y. Let q denote her belief over F_R, i.e. her (joint) belief over these three parameters. Her expected utility is given by:

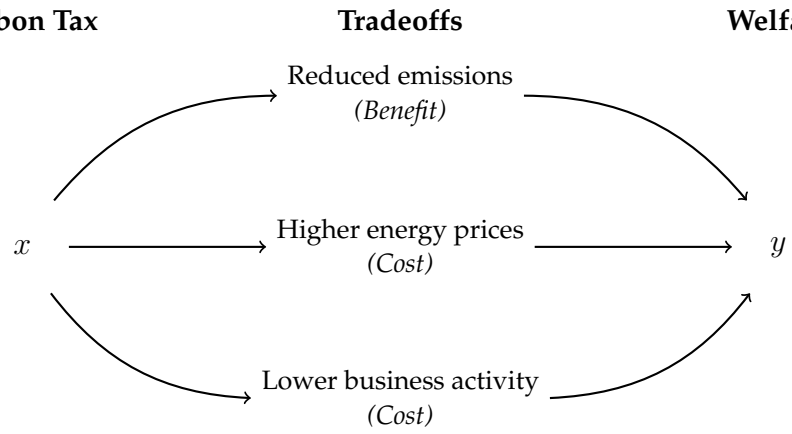
$$U_q(x) = \mathbb{E}_q[f_R(x)].$$

Her goal is to choose x to maximize $U_q(x)$.

In our running example, R is a policymaker who designs a carbon tax x. y represents the welfare of a representative household or voter. Higher taxes reduce emissions and improve health outcomes, but may also increase energy prices and reduce private sector activity. The single choice of a tax rate x may have different consequences for welfare, depending on the *aggregate* of these different effects. The policymaker would like to balance these effects and choose the tax rate that maximizes expected welfare. Figure 2 illustrates this idea below.

[1] I assume a quadratic form for tractability but show that the central intuitions generalize in Appendix B.5.

Figure 2: Effect of x on y through tradeoffs: example of policymaker and consultant



However, there are many other interpretations of the receiver and her preferences. If R is an investor, she cares about how much money she can make by investing in a choice x. If R is a CEO restructuring her workforce, she cares about the effect of layoffs on profits. If R is a voter, she cares about the impact of a policy x on her wellbeing.[2]

2.2 Sender and Data

The receiver R contracts a sender S to provide her with information about the relationship between x and y. This sender should be thought of as a *consultant* or *contractor* who provides domain expertise and access to data. Of particular interest is also the case where S is an AI program, such as a large language model (LLM) that R can interface with.

The sender may consider *simpler* models than the receiver. The form of his simplicity is determined by a private type θ. The order of events is as follows.

1. Nature draws S's type and the receiver's true model f_R.
2. S collects data. These data are not observable to R.
3. S provides a model to R that fits the data he collected. R updates her beliefs about the true data-generating process based on her beliefs about S's type and the model.
4. R makes a decision x^*. S receives utility. Uncertainty resolves, and R receives her utility.

I now provide more detail about each of these components.

[2] I utilize these simple preferences for higher y in the baseline model, but Section 4.2 extends them to evaluate settings where R also values *forecast accuracy*, and thus knowing the correct model of the world.

Sender's Type A sender's private type θ affects the *simplicity* of the models he is capable of providing to R. We denote the type space by Θ. There are two varieties of sender. The first is an **accurate** sender. We denote his type a. This sender provides a model in the set

$$F_a := \{f_a \in F_R : f_a(x) = \beta_0 + \beta_1 x - \delta x^2, (\beta_0, \beta_1) \in \mathbb{R}^2, \delta \in \Gamma\}.$$

We note that $F_a = F_R$, i.e. the accurate sender is estimating *exactly* the same models as R. Additionally, we assume he has the same prior over models as the receiver. The ex-ante probability of seeing a type a sender is p_a.

The second variety of sender is a **simple** sender. This sender has a type δ, where $\delta \in \Gamma$. A simple sender is limited in how he can estimate the *curvature* of models. [3] A type δ sender is able to provide R a model in the set F_δ.

$$F_\delta := \{f_\delta \in F_R : f_\delta(x) = \beta_0 + \beta_1 x - \delta x^2, (\beta_0, \beta_1) \in \mathbb{R}^2\}.$$

In particular, I assume that a simple, type δ sender **calibrates** the curvature of the convex loss to δ, and does not estimate it on the data. For example, the consultant may calibrate the effect of carbon taxes on energy prices using a fixed elasticity from the literature, rather than estimating it on data that may be more or less relevant to the receiver's setting. Or, he may believe that the cost of actions is always given by δx^2, based on his own idiosyncratic experiences.

I assume a type δ sender has a prior over β_0 and β_1 given by measures $\beta_0 \sim P_0^\delta$ and $\beta_1 \sim P_1^\delta$ with support on $\mathbb{R}$. We denote by $P_{f_S}^\delta$ a type δ sender's prior over models f_S. His second-order beliefs about R's beliefs are as follows: he behaves as if R's prior over α_0 and α_1 are flat, but correctly believes her prior over γ is given by p_γ^R. [4] The simple sender thus estimates *misspecified models*; he truthfully believes that his way of modeling the problem (in simple terms) is the right way, and may "agree to disagree" with a receiver who would entertain a richer model.

The receiver sees a simple sender with ex-ante probability $1 - p_a$. A simple sender's type is in turn drawn from a distribution T over $[0, \overline{\gamma}]$ known to R. If T is continuous with pdf t, the ex-ante density of a type δ simple sender is thus $(1 - p_a)t(\delta)$. In turn, the full type space Θ is given by $\Theta = \Gamma \cup \{a\}$.

Two Simple Types To further motivate the idea of simple senders, I introduce a running "two-simple-sender type" example. The type $\delta = 0$ sender estimates a model $y = \beta_0 + \beta_1 x$, without allowing for any nonlinear effects of x. This sender always understates the curvature of the true DGP (he assumes it is 0). In this sense, he **neglects tradeoffs** in that he ignores any curvature in the true DGP: he runs a linear regression of y on x instead of considering a nonlinear relationship.

[3] Appendix B relaxes this assumption and allows types to be intervals of curvature values, so that a sender may neglect or exaggerate some tradeoffs while correctly estimating others. The main results are robust to this richer type space. However, the design problem studied in Section 5 shows that if a receiver saw a simple sender in equilibrium, she would only see one of these *singleton* types.

[4] This is a regularity assumption that shuts down edge-cases of learning in equilibrium and, in particular, variations in the receiver's expectation of γ with respect to certain moments of the distribution.

In our example, the policymaker may believe small carbon taxes generate meaningful reductions in pollution without adverse effects, but very high taxes lead to higher energy prices or lower business activity. These nonlinear effects would be neglected if a consultant presented a linear regression of welfare on tax rates.

The type $\delta = \overline{\gamma}$ sender is the opposite. He estimates models of the form $y = \beta_0 + \beta_1 x - \overline{\gamma} x^2$, where $\overline{\gamma}$ is fixed at its highest possible value. This sender always overstates the curvature of the true DGP and **exaggerates tradeoffs**. This reflects a situation where a sender estimates a "worst-case scenario." For example, he may assume that higher taxes always have maximally deleterious effects on the business sector, which the receiver may believe is unlikely.

Sender's Data Prior to choosing a model, S chooses a data sampling strategy given by μ, where μ is a distribution over $\mathbb{R}$. R does not observe S's choice of μ. His choice set of distributions is given by M, which has the following properties.

Assumption 1. *M is given by the set of symmetric distributions on $\mathbb{R}$ such that for all $\mu \in M$:*

1. $\sigma_\mu^2 := \mathrm{Var}_\mu(x) \in [\underline{\sigma}^2, \overline{\sigma}^2]$ *for $\underline{\sigma}$ arbitrarily small and $\overline{\sigma}$ sufficiently large;*
2. *and* $x_\mu := \mathbb{E}_\mu[x] \in X$.

Assumption 1 allows the sender to choose any symmetric distribution over $\mathbb{R}$ such that its variance lies in a compact interval, and such that its mean lies in the receiver's choice set X.[5] We will refer to a type θ sender's choice of μ as μ_θ, and denote its mean x_θ and variance σ_θ^2. I note that μ_θ does allow the sender to sample information about x values outside of X. While I make this assumption for tractability, it is also consistent with the interpretation that the receiver's choices are "bounded" in a broader space of potential choices, and that choices outside of X are concretely "irrelevant" to the receiver's eventual decision. This latter interpretation will be particularly important when we analyze the sender's choice of σ_θ.

We introduce two definitions here. We say a type θ sender is a **generalist** if $\sigma_\theta = \overline{\sigma}$. Intuitively, a type θ sender's model is estimated *distantly* on as wide a range of x values as possible. We say a type θ sender is a **specialist** if $\sigma_\theta = \underline{\sigma}$. That is, this sender estimates a model *locally* on a small set of x values. In our example, when the consultant is first contracted by the policymaker, he will choose whether to position himself as a *generalist* or *specialist*. If he is a generalist, then the data he uses to fit his models is drawn from a wide range of carbon tax rates from many contexts. If the consultant is a *specialist*, he will utilize information about a small range of carbon tax rates.

Given this sampling strategy, S directly observes the true joint distribution of (y, x), subject to observational noise. Formally, if the true data-generating process is f_R^*, he observes a joint distribution $D(y, x)$, where the marginal distribution of x is given by $D_x = \mu_\theta$ and the conditional

[5] The assumption of symmetry is used for tractability to discipline the set of models the receiver will entertain, but the results generalize. Relatedly, the constraint that $\sigma > 0$ is a regularity assumption made to ensure learning is structured in equilibrium.

distribution of y given x is given by $D_y(y|x) = f_R^*(x) + \eta$, where η is a mean-zero noise term with unknown variance. [6]

Sender's Information With the sender's type and sampling strategy in hand, we define the model S provides to R. We say that a model f_S **best-fits the data** with respect to θ and a data-sampling strategy μ_θ if:

$$f_S = \arg\min_{f \in F_\theta} \mathbb{E}_{D(\mu_\theta)}\big[\big(y - f(x)\big)^2\big].$$

For example, when $\theta = \delta = 0$, f_S represents the *best-fitting linear model* to the data.[7]

I assume S always provides R the best-fitting model to the data. In our running example, the interpretation is that the consultant makes an honest effort to provide R quality information about tax rates, within the constraints of how he approaches the problem. S may use a well-known algorithm to fit his model (such as a linear regression).[8] This allows me to isolate how model *simplicity* and strategic *data collection* shape learning and sender behavior.

Sender's Preferences Suppose S provides a model $f_S \in F_\theta$. R has a belief over both other plausible models as a function of f_S, her beliefs about the data μ she believes S used, and her belief about S's type. Let x^* be the receiver's choice of $x \in X$ conditional on her posterior beliefs over models q given f_S and her beliefs about μ. A type θ sender's utility is then:

$$U^\theta(x^*) = \mathbb{E}^R_{q|f_S}[f_R(x^*)].$$

Conditional on providing a model f_S, S would like to maximize R's posterior expected utility at her chosen action.

The interpretation of these preferences in our example is that the consultant would like to make the policymaker *optimistic*. He may receive a consulting bonus if she is more optimistic about the potential gains she can reap from the carbon tax policy. Or, if the policymaker is more optimistic about the effects of the carbon tax, he is more likely to adopt some new policy, which the consultant is able to take credit for assisting with. Beyond the current example, however, the preferences lend themselves to several interpretations.

- S is a portfolio manager who receives a fixed share of R's expected earnings. He cares about the expected profits R can realize (and thus his payment).
- S is a politician who provides information about y to a median voter R. The higher the median voter thinks y is, the more likely she is to turn out in an election.

[6] I allow S to observe the distribution of data to shut down the role of idiosyncratic data draws in shaping posterior beliefs. This is equivalent to S observing an arbitrarily large (or infinite) number of i.i.d. draws from the true data-generating process.

[7] The existence of a best-fitting model is given by the Hilbert Projection Theorem.

[8] The policymaker could at least partially verify this through a "replication package."

- S is an AI program that wants to maximize its engagement with its user base. The more optimistic users are, the more likely they will be to purchase a premium subscription.

Equilibrium Finally, I formally define equilibrium for the sender and receiver. Given the probability p_a of an accurate sender and the distribution T of simple sender types, an **equilibrium** consists of a sampling strategy $\mu_\theta^* \in M$ for each type $\theta \in \Theta$, a belief $\hat{\mu}_\theta \in M$ about each type's sampling strategy, a posterior belief $q(f_R|f_S)$ over F_R for each model f_S, and a choice $x^*(f_S) \in X$ for each model f_S, such that the following are satisfied.

1. A type θ sender who samples according to μ_θ^* provides the model $f_S \in F_\theta$ that best-fits the data with respect to θ and μ_θ^*.
2. For each f_S, $q(f_R|f_S)$ is derived by Bayes' rule from R's prior over f_R, the probability p_a, the type distribution T, and her belief $\hat{\mu}_\theta$ about the sampling strategy of each type $\theta \in \Theta$.
3. For each f_S, $x^*(f_S) \in \arg\max_{x \in X} \mathbb{E}_{q|f_S}[f_R(x)]$.
4. For each $\theta \in \Theta$,

$$\mu_\theta^* \in \arg\max_{\mu \in M} \mathbb{E}_{f_S}^{\theta}\Big[\max_{x \in X} \mathbb{E}_{\hat{q}(f_R|f_S;\mu)}[f_R(x)]\Big],$$

   where $\hat{q}(f_R|f_S;\mu)$ is S's second order belief about R's posterior given f_S, when R believes S sampled according to μ.
5. For each $\theta \in \Theta$, $\hat{\mu}_\theta = \mu_\theta^*$, i.e. the receiver's beliefs over μ_θ are equal to their actual distribution.

Because R does not observe μ_θ, a sender's expected utility depends only on R's belief $\hat{\mu}_\theta$ about his sampling strategy, and any sampling strategy that R correctly anticipates is a best response. Condition 4 selects, among these, the sender-preferred equilibrium: each type samples in the way that maximizes his expected utility when R correctly anticipates it.

2.3 Additional Comments

In this framework, the receiver's learning is affected by two key frictions. The first is that the sender's models are *simple*: S may be unable to think about problems exactly the way R would like. In the two-simple-type example, S either estimates a *naive* linear model that *totally* neglects tradeoffs, or a *cautious* model that totally exaggerates them and assumes a worst-case scenario. This assumption can be founded through a few different lenses.

1. **Lack of sophistication**: the case where S estimates a naive model is equivalent to one where S estimates a *linear regression* of y on x. The sender may be able to estimate a "line of best fit," but not understand more complicated model fitting techniques. Thus, he does one of two things. Either he continues to totally neglect nonlinear tradeoffs (because he cannot estimate them), or he overcompensates and assumes a worst-case scenario.

2. **Calibration**: instead of estimating parameters organically on data, S may calibrate his models' curvature to a fixed value, such as an elasticity from the literature.

3. **Strategic Behavior**: the sender may be able to make the receiver *more optimistic* if she thinks his model was a simplified model. This will be a central result of the paper (Section 5).

The second friction is that the receiver does not observe the sender's underlying data and must infer it. For example:

1. The sender may have access to a proprietary dataset, which cannot be directly disclosed to the receiver. Commonly used datasets like those from Nielsen provide proprietary information on "audience segments," for instance, to help advertisers run targeted campaigns (Nielsen, 2024).

2. The sender's model may also embed *expertise*. The strategy literature suggests that firms often develop internal ways of approaching a problem based on analogous situations they have experienced in the past (Carroll and Sørensen, 2024). Accordingly, a consulting firm's recommendations may be informed by experience working on cases similar to each other (as a "specialist") or drawn from many different cases and past contexts (as a "generalist").

3. The sender is an AI program. It is often difficult for users of these programs (or even their developers) to understand the mapping from internal data to recommendations (Zuccarelli, 2024). In this sense, the mapping from the data to a model is a *black box*.

What matters is that R knows S has access to useful data, but must infer how that data may be strategically sampled when estimating a model.[9]

Additionally, R may also be unable (or unwilling) to expend organizational time and resources in collecting, understanding, and synthesizing complex data. Instead, she delegates the collection of data and its synthesis into a model to the sender, expediting her decision-making process in the interest of convenience, but potentially at the cost of information quality.

Finally, I assume that S must provide the best-fitting model to the data, rather than a model that simply "fits the data." The interpretation is that S makes an honest effort to provide R the best model he can, given the set of data-generating processes he can model. While it could be that S's honesty emerges due to professional norms or unmodeled career concerns, it may also be that S synthesizes data using methods or algorithms that are well-known in his field of expertise. [10] Nevertheless, even though S provides a model of best-fit, he can still manipulate the set of models R entertains through the way he initially samples data as well as his type δ.

I focus largely on settings where the nonlinear component of f_R is quadratic. While this is done for tractability, Appendix B.5 discusses the assumptions necessary to generalize the logic.

[9] Accordingly, I will be interested in exactly *what sorts of data* the sender will collect in equilibrium, and *how* that data collection interacts with the simplicity of his models.

[10] Relatedly, R could require S's methodology to be partially "replicable."

3 Analysis

This section first characterizes the set of models that the receiver entertains, and then shows how senders strategically sample data. I show that senders who neglect tradeoffs position themselves as "generalists," while those who exaggerate them position themselves as "specialists." In the two-simple-type example, the simple sender who provides a "linear regression" model neglects tradeoffs, while the sender who provides a "worst-case scenario" exaggerates tradeoffs. I show how this result emerges from the combination of senders' model simplicity with their ability to strategically sample data.

In either case, I show that the simple sender's model is estimated on data that will ex-ante likely be *irrelevant* to the receiver's eventual choice, based on *where* he samples. When a simple sender samples as a generalist, he samples in the center of the receiver's choice space, but includes thick-tailed data about choices at the *edge* of her choice space. If the simple sender samples as a specialist, he only samples data in a small neighborhood on one of the edges of the receiver's choice space.[11]

I then further generalize the baseline model with *multidimensional choices*. This delivers an additional prediction: senders may sample data to maximize *collinearity* between different variables. In particular, if a sender *neglects* interaction effects between two variables, he samples data so that they are as *correlated* as possible. If he *exaggerates* interaction effects, he samples data so that they are as *anticorrelated* as possible. Thus, although receivers correctly infer the sort of data that senders collect in equilibrium, that data may be *low quality*, in that it suffers from a high degree of collinearity, and embeds information about choices that will (in expectation) be irrelevant.

3.1 Receiver's Learning and Behavior

I begin by characterizing receiver learning as a function of data-sampling strategies and S's type.

Receiver's Updating Suppose S provided a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ which best fits the data. If the sender is an accurate sender, this is the true model. However, if the sender is a simple sender, this may not be the true model. We define $\phi(f_S; \mu_\delta, \mu_a)$ as the set of functions that could rationalize the data for R, given f_S and both a type δ and type a's sampling strategy.

$$\phi(f_S; \mu_\delta, \mu_a) = \left\{ f_R \in F_R : f_S = \arg\min_{f \in F_\theta} \mathbb{E}_{\mu_\theta}\left[\left(f_R(x) - f(x)\right)^2\right], \theta \in \{a, \delta\} \right\}.$$

If $\theta = a$, then because $F_a = F_R$, the best-fitting model is f_R itself, so that $f_R = f_S$. If $\theta = \delta$, we necessarily have that for each such f_R:

$$\frac{\partial}{\partial \beta_i} \mathbb{E}_{\mu_\delta}\left[\left(f_R(x) - f(x)\right)^2\right] = 0,$$

[11] Appendix B generalizes the results to senders who may only neglect or exaggerate some tradeoffs while correctly estimating others.

which yields the following moment conditions familiar from ordinary-least-squares (OLS):

$$\mathbb{E}_{\mu_\delta}\left[f_R(x) - \beta_0 - \beta_1 x + \delta x^2\right] = 0 \tag{1}$$

$$\mathbb{E}_{\mu_\delta}\left[x\left(f_R(x) - \beta_0 - \beta_1 x + \delta x^2\right)\right] = 0 \tag{2}$$

Equation (1) is the moment condition with respect to the intercept, and states that the difference between f_R and f_S must average to zero. We note that this is the same as the constraint that a model must "fit the data." Equation (2) can be combined with equation (1) to read that the covariance between x and $f_R(x) - \beta_0 - \beta_1 x + \delta x^2$ is zero. Both expectations are taken with respect to μ_δ, which is the type δ sender's sampling strategy. The following proposition characterizes the set of functions that R entertains given f_S, δ, and μ_δ: i.e., assuming the receiver *knows* the simple type δ sender's underlying data sampling strategy.

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and the simple sender's sampling strategy μ_δ has mean and variance x_δ and σ_δ^2. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + (\gamma - \delta)\left(\sigma_\delta^2 - (x - x_\delta)^2\right).$$

The proof of this result is contained in Appendix A. Consider the policymaker and consultant. The consultant's model may be accurate. In this case, $f_R(x) = f_S(x)$, and the consultant's model is the true model. However, it is also possible that the sender's model was simple. Thus, she considers both the consultant's original model and other models that could rationalize the tax-welfare data the consultant saw.

In particular, note that the set of models she considers depends *only* on the sampling strategy of the type δ sender, not the type a sender: the accurate sender's model f_S is itself the member of this set with $\gamma = \delta$. Mathematically, the set of models she considers, given a best-fitting model f_S, takes a simple form: they are all the sender's original model plus a constant, less a quadratic term centered at x_δ. In the case where $\delta = 0$, we have

$$f_R(x) = \beta_0 + \beta_1 x + \gamma\left(\sigma_\delta^2 - (x - x_\delta)^2\right)$$

for each $\gamma \in [0, \overline{\gamma}]$. In the case where $\delta = \overline{\gamma}$, we have

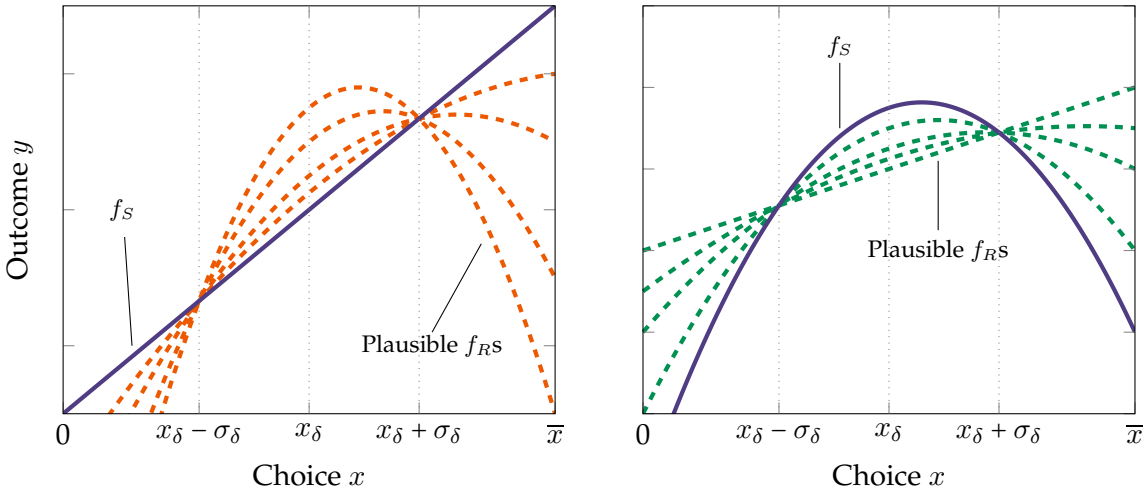
$$f_R(x) = \beta_0 + \beta_1 x - \overline{\gamma} x^2 + (\gamma - \overline{\gamma})\left(\sigma_\delta^2 - (x - x_\delta)^2\right)$$

for each $\gamma \in [0, \overline{\gamma}]$. In both cases, the set of models R considers involves a fixed "affine component" plus a constant and quadratic term, which may be positive or negative. Moreover, all plausible f_Rs have the same slope at x_δ.

This intuition is shown for the type $\delta = 0$ in the left panel of Figure 3 below. The purple line shows the sender's linear regression model. The orange dashed lines indicate all the potential alternative models f_R that the receiver entertains. When the type $\delta = 0$ sender estimates a linear

regression model, he neglects any tradeoffs R thinks are important, and thus estimates a relatively naive model. Accordingly, all the additional models that R entertains have a steeper curvature than the regression model, precisely because the linear regression neglects tradeoffs. However, across the sampling range, they have the same sample mean and average slope.

Figure 3: Plausible receiver models f_R as a function of sender model f_S and sampling strategy μ_δ



(a) Plausible models when sender provides naive "linear" model

(b) Plausible models when sender provides cautious "worst-case" model

The right panel shows the intuition for the type $\delta = \overline{\gamma}$ sender, who estimates a cautious model that exaggerates tradeoffs. Here, all the additional models that R entertains have shallower curvature than f_S. Just like the case with the linear regression, however, they all have the same average slope and sample mean with respect to μ_δ.

Receiver's Decisionmaking The receiver's learning takes a relatively simple form: regardless of the limitation on S's ability, R always learns what the data-generating process looks like "up to the first derivative at x_δ and a constant." Moreover, as σ changes — and thus S's sampling strategy changes — the set of plausible f_Rs remains identical, except that they shift up or down by a constant. This means that R's optimal *action* depends only on the probability $\Pr(\theta = a|f_S)$ that the model is accurate, her expectation of γ given that the model is simple, and the sampling choice of a type δ simple sender (and not the accurate sender). Given f_S, this choice has the following expression.

Corollary 1. *Given $f_S = \beta_0 + \beta_1 x - \delta x^2$ and a type δ sender's sampling strategy μ_δ, R's optimal action is*

given by $x^*(f_S)$, where:

$$x^*(f_S) = \max\{0, \min\{\tilde{x}, \overline{x}\}\}$$
$$\tilde{x} = x_\delta + \frac{f_S'(x_\delta)}{2\Big(\delta + \big(1 - \Pr(\theta = a|f_S)\big)\big(\mathbb{E}^R_\gamma[\gamma|f_S, \theta \neq a] - \delta\big)\Big)}.$$

We note that $\tilde{x}$ is simply R's optimal interior action, and that $x^*(f_S)$ is simply this interior choice truncated between 0 and $\overline{x}$.

3.2 Sender's Data Sampling and Model Provision

I use these insights to show how S chooses the *spread* of his data as a function of his type. First, we introduce the following notation:

$$\Delta(f_S) = \mathbb{E}^R_\gamma[\gamma|f_S, \theta \neq a] - \delta.$$

Fixing f_S, Δ represents the *difference* between R's expectation of the quadratic parameter γ, conditional on the model being simple, and the sender's provided parameter δ. In turn, given a model f_S, R's expected utility is

$$\mathbb{E}^R_{q|f_S}[f_R(x)] = f_S(x) + \big(1 - \Pr(\theta = a|f_S)\big)\Delta(f_S)\big(\sigma^2_\delta - (x - x_\delta)^2\big).$$

We say a type δ simple sender estimates a **naive model** if $\mathbb{E}^\delta_{f_S}[\Delta(f_S)] > 0$, and a **cautious model** if $\mathbb{E}^\delta_{f_S}[\Delta(f_S)] < 0$. That is, a naive model sender tends to *neglect* tradeoffs, underestimating γ relative to what the receiver considers, while a cautious model sender tends to *exaggerate* them. Concretely, a type δ sender estimates a naive model if $\delta < \mathbb{E}^R_\gamma[\gamma]$ and a cautious model if $\delta > \mathbb{E}^R_\gamma[\gamma]$. In particular, the type $\delta = 0$ sender's linear regression is a naive model, and the type $\delta = \overline{\gamma}$ sender's worst-case model is a cautious model. These definitions allow me to state my main result on strategic sender sampling.

Theorem 1. *If a type δ simple sender S estimates naive models, he positions himself as a generalist ($\sigma^*_\delta = \overline{\sigma}$) and sets $x^*_\delta = \mathbb{E}^\delta_{f_S}[x^*(f_S)]$, where $x^*(f_S)$ is the sender's expectation of the receiver's choice given f_S. If he estimates cautious models, he positions himself as a specialist ($\sigma^*_\delta = \underline{\sigma}$) and sets $x^*_\delta = 0$ or $\overline{x}$.*

Theorem 1 can be interpreted as follows. If the sender estimates a naive model, such as the type $\delta = 0$ sender's linear regression, the receiver will consider models that have *steeper curvature* than the original model on average. This creates an incentive for him to position himself as a *generalist*. In particular, his equilibrium choice of distribution μ^*_δ is one such that x^*_δ is in the center of the receiver's choice space, precisely where he expects the receiver to make her choice, and where $\sigma^*_\delta = \overline{\sigma}$ is as high as possible. This distribution is thus as spread out as possible.

However, if the sender estimates a cautious model, such as the type $\delta = \overline{\gamma}$ sender's worst-case model, the receiver will consider models that have *flatter curvature* than f_S on average. This gives

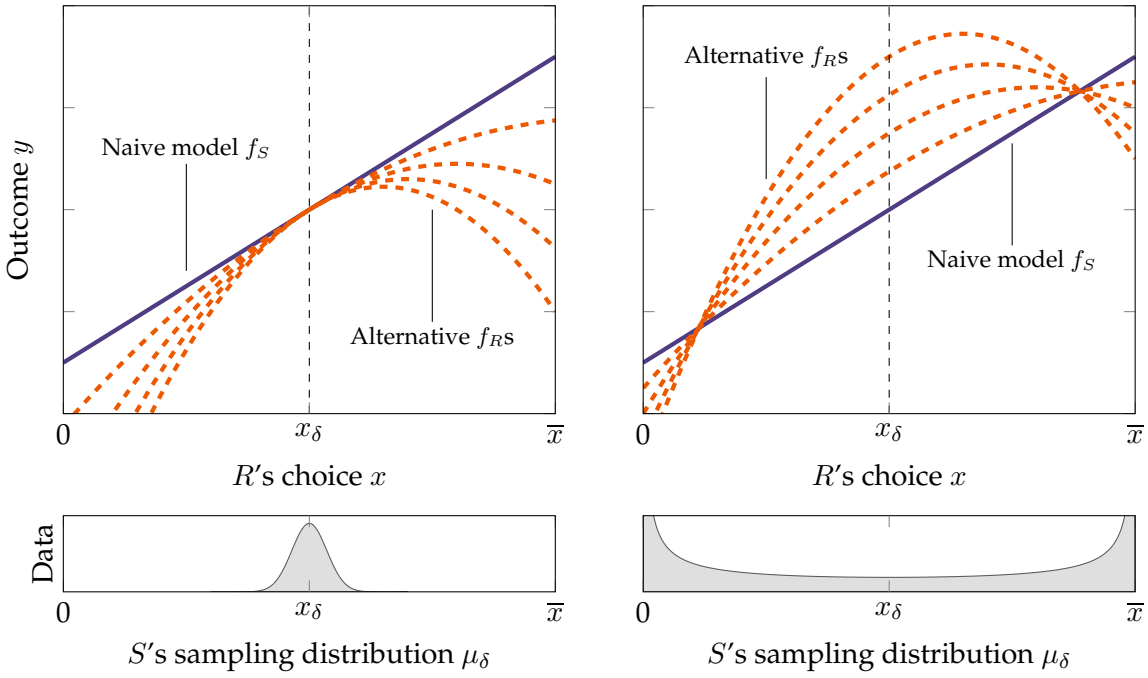
S an incentive to position himself as a specialist who estimates the DGP locally in a neighborhood of a point. The theorem shows that this sender will sample data *locally* about either $\overline{x}$ or 0; his equilibrium choice of μ^*_δ is such that x^*_δ is either $\overline{x}$ or 0 and $\sigma^*_\delta = \underline{\sigma}$ is as low as possible.

The intuition is driven by the interaction of the curvature of the receiver's additional models with the sender's method of *data sampling*, and is shown visually in Figure 4 below. The top panels (a) and (b) look at the case where f_S is a naive model linear regression of y on x. The bottom panels (c) and (d) look at the case where f_S is the worst-case model fitting the data. The smaller graphs below the graphs of f_S and f_R show what the distribution of *data* underlying different choices is.

In panel (a), a naive S has positioned himself as a *specialist*, and samples data locally in a neighborhood of x_δ, shown by the distribution of data in the smaller graph below. All the additional plausible models f_R are tangent to f_S at x_δ. These alternative models have steeper curvature (i.e. a higher γ value) than f_S, and lie below f_S at the point of tangency. This logic flips when looking at panel (b), where S has positioned himself as a generalist. Accordingly, the smaller graph below shows that the distribution of data has *spread out* around x_δ. Recall that the distribution of x values when $\sigma = \overline{\sigma}$ is a mean-preserving spread of that when $\sigma = \underline{\sigma}$. To rationalize the data, functions with higher γ values that are "more concave" must shift up relative to shallower ones. In panel (b), this means that the steeper f_R models now lie *above* f_S.

Figure 4: Models that R entertains when S estimates naive vs. cautious models, specialist vs. generalist

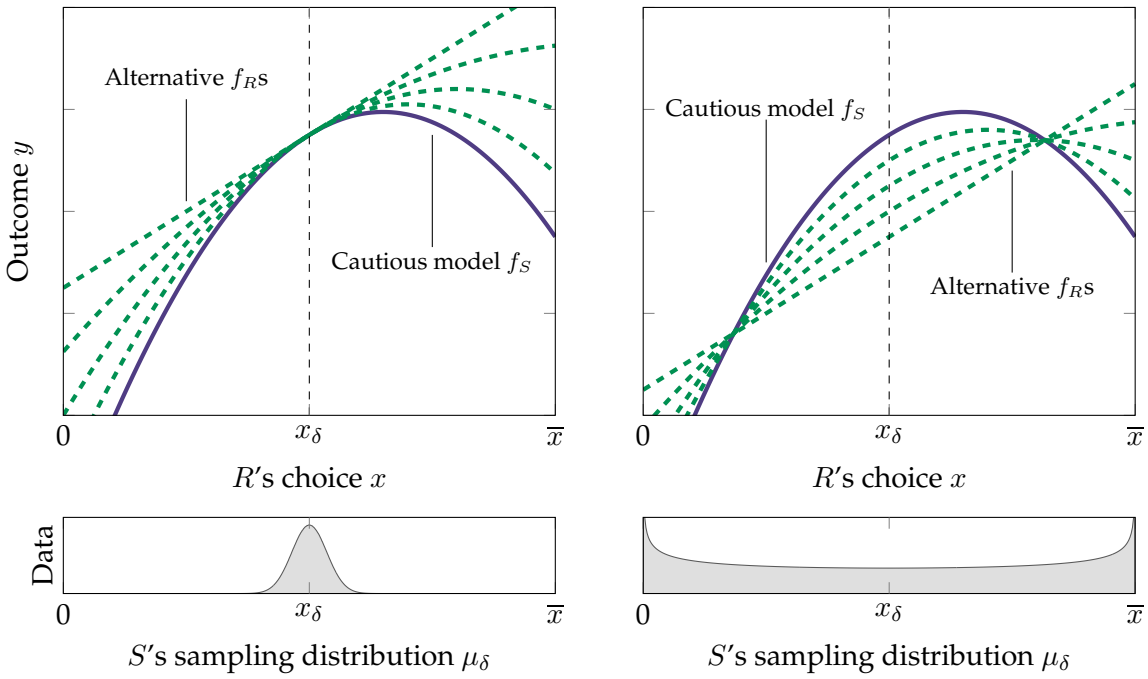
(i) S estimates naive "linear" model



(a) S is a specialist

(b) S is a generalist

(ii) S estimates cautious "worst-case" model



(c) S is a specialist

(d) S is a generalist

The higher the alternative f_Rs are above f_S, the more optimistic R is about her prospects.

The logic is flipped in panels (c) and (d). When the sender estimates a "worst-case" model, all the additional models R entertains have shallower curvature (i.e. lower γ) than the sender's model. When he samples as a specialist, all these additional models lie *above* the original model. When he samples as a generalist, these models lie *below* the original model. Thus:

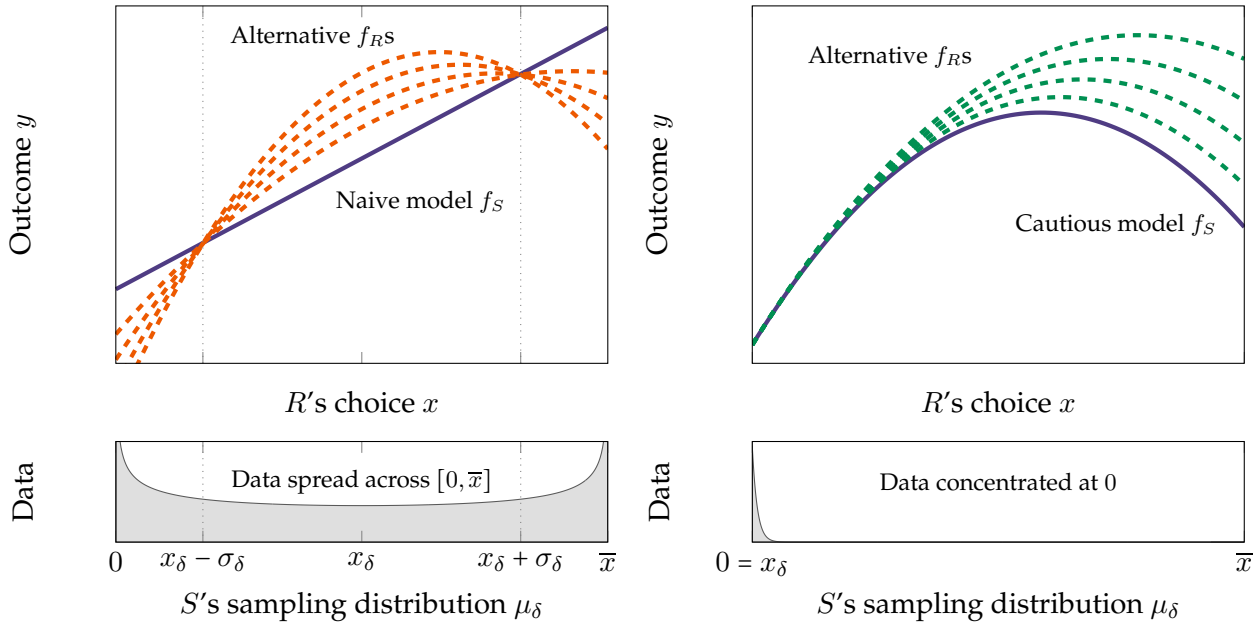
- When S presents naive models, he gains the most by positioning himself as a *generalist*.
- When S presents cautious models, he gains the most by positioning himself as a *specialist*.

Theorem 1 additionally shows the value of x^*_δ a type δ sender will choose. The intuition is as follows. When presenting a naive model, S would like to ensure all the additional models that R entertains are as far above f_S as possible. This gap is largest when x_δ is at the maximum of f_S, i.e. where $f'_S(x_\delta) = 0$. Because S does not know ex-ante where this maximum will be, he thus samples where he *expects* f_S to be the shallowest.

When S estimates a cautious model, the logic is the opposite. If f_S is maximized at a point x, so are all other models tangent to f_S. However, if all these other models are tangent at a point farther from x, the set of tangent models grows higher and higher above f_S. Thus, he samples at either 0 or $\overline{x}$, where he expects f_S to be steepest.

This is visualized in Figure 5 below. The left shows a naive (linear regression) sender positioning himself as a generalist who samples widely around a policy x_δ in the center of the choice space. The right shows a cautious (worst-case-scenario) sender positioning himself as a specialist near 0.

Figure 5: Models receiver entertains in equilibrium and data naive vs. cautious sender collects



(a) Naive S samples as a generalist

(b) Cautious S samples as a specialist

In general, the sender samples data in a way that is *irrelevant* for the receiver's eventual choice. Even though the naive sender's data is centered between 0 and $\overline{x}$, he conditionally makes the receiver more and more optimistic the larger the variance of the data he samples. In practice, this means that he prefers a bimodal distribution with modes close to 0 and $\overline{x}$, rather than a more even single-peaked distribution in the center of the receiver's choice space. This can also be seen visually: the sender's model is *most accurate* at $x_\delta + \sigma_\delta$ and $x_\delta - \sigma_\delta$ — given by the fact that all the receiver's alternative models are equal to f_S at these points. However, $x_\delta \pm \sigma_\delta$ grow arbitrarily large or small with σ_δ. Thus, the naive sender's data suffers from *excessive variance*.

The cautious sender creates the same issue by sampling in the smallest possible neighborhood of 0 or $\overline{x}$. In this sense, he accurately describes the expected value of y given x at just one of these extreme points, but nowhere else. In the example of the policymaker and consultant, this means that a consultant would mostly collect data where a carbon tax is close to 0, or where a carbon tax is excessively high, neither of which is likely to represent the policymaker's optimum.

The coupling of *model simplicity* with *strategic sampling* thus creates incentives to collect *poor data* about the effects of choices on outcomes. Crucially, I note that the receiver, in equilibrium, is not uncertain as to the data the sender collected, and in fact *perfectly* infers what that data are. However, the informational content of those data is ultimately lower than she would like. Section 4.2 formally shows that the *second-best* degree of sampling would be an interior amount between $\underline{\sigma}$ and $\overline{\sigma}$.

The naive linear regression model (type $\delta = 0$ sender) is also of note. The result on strategic sampling says that this sender always estimates models using data sampled from widely across the choice space. However, regressions are most effective when thought of as *local linear approximations*. Thus, the result on sampling contrasts a sender's persuasive desire to fit a regression to "distantly sampled data" with the common perception that regressions are most effective when utilized as "local" linear approximations.

Accurate Sender's Sampling and Utility We note that because the accurate sender's models are not misspecified, his data sampling strategy does not affect the alternative models R considers. Any sampling strategy is therefore optimal for him and irrelevant to R's beliefs; in equilibrium, I assume he adopts an identical sampling strategy to the type $\delta = 0$ simple sender.

We note that the accurate sender objectively *gains* from the existence of the simple sender, in the sense that he can make the receiver more objective than the truth. Concretely, suppose the accurate sender provides a model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. Because the receiver believes that this model was, with some probability, provided by a simple sender, he also entertains additional models. These additional models deliver higher maximal expected utility than $f_S(x)$.

However, the accurate sender knows he has provided the true model. From his perspective, the existence of the simple senders makes the receiver *more optimistic than the truth*. The magnitude

of this gain is given by:

$$\max_x f_S(x) + \big(1 - \Pr(\theta = a|f_S)\big)\Delta(f_S)\big(\sigma_\delta^2 - (x - x_\delta)^2\big) - \max_x f_S(x),$$

where we note that $\Delta(f_S)\big(\sigma_\delta^2 - (x - x_\delta)^2\big)$ is always larger than 0 by naive/cautious senders' forms of strategic sampling. Thus, the accurate sender gains from the existence of simple senders. Moreover, it is precisely these senders' simplicity in considering *curvature* coupled with *strategic sampling* that permits this gain. His ex-post gain is larger the more likely he is to be perceived as a simple sender. We probe this result further in Section 5, when we characterize what optimal *composition* of senders advantages the accurate sender the mosts.

Interval Types We have thus far assumed that simple senders' types are *singletons*, i.e. they assume models have a fixed curvature δ. However, it is plausible that a sender neglects some tradeoffs and exaggerates others. We could indeed imagine a sender who fits models of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$, but where δ can take any value in an *interval* $\theta \subsetneq \Gamma$. If $\theta = [0, \frac{\overline{\gamma}}{2}]$, for instance, this sender provides the correct model if the true γ is less than $\frac{\overline{\gamma}}{2}$, but simplifies the model if the true γ exceeds $\frac{\overline{\gamma}}{2}$.

Appendix B extends the baseline model with interval types and shows that the result generalizes. The key is that $\Delta(f_S)$ can potentially take different values. Suppose the sender can fit an interval θ of δ values. If δ is in the interior of θ, then $\Delta(f_S) = 0$. If δ is at the lower boundary, then $\Delta(f_S) \leq 0$. If δ is at the upper boundary, then $\Delta(f_S) \geq 0$. The type δ simple sender then integrates over these different possibilities and choose μ_δ accordingly.

3.3 Multidimensional Choices

While the baseline model considered a single choice variable x, I also consider what happens when choices are multidimensional. For example, imagine that our policymaker can shape consumer welfare not just through a carbon tax on businesses but also a complementary clean energy subsidy. I briefly show that many of the findings of the baseline model generalize. However, I also show that when the sender samples data about multiple variables, he always has an incentive to introduce nearly perfect *collinearity* between them. In particular, if there are two choice variables — x_1 and x_2 — and the sender can choose how *correlated* the two variables are when sampling the data, he will make them either as correlated as possible or as *anticorrelated* as possible.

Multidimensional Models Suppose now that the receiver's choices are given by $x = (x_1, \ldots, x_n) \in X^n \subseteq \mathbb{R}^n$. The models f_R the receiver entertains are now as follows. For each $f_R \in F_R$, we have

$$f_R(x) = \alpha_0 + \alpha^T x - x^T \gamma x,$$

where $\alpha_0 \sim P_0^R$, $\alpha \sim P_1^R$ for measures on $\mathbb{R}$ and $\mathbb{R}^n$, and $\gamma \sim P_\gamma^R$ on a set Γ. Γ is a compact, convex set of positive-semidefinite matrices such that $0 \in \Gamma$ and such that there exists $\gamma \in \Gamma$ with $\gamma_{ii} > 0$ for

all i.

F_R now *generalizes* the idea of a quadratic data-generating process to a multivariate setting using a set of concave *quadratic forms*. This set of forms is given by a set of positive-semidefinite matrices Γ. Accordingly, I assume the type space Θ of senders is given by $\{a\} \cup \Gamma$.

For simple senders, I assume that A type δ sender provides models of the form:

$$f_S(x) = \beta_0 + \beta^T x - x^T \delta x,$$

where $\delta \in \Gamma$ is the sender's type.

Finally, I describe how the sender samples data in multiple variables. Formally, let $\epsilon \in \mathbb{R}^n$ be a *spherically symmetric* random vector that admits a density with $\mathbb{E}[\epsilon] = 0$ and $\mathrm{Cov}(\epsilon) = I_n$. Let Σ_δ be a symmetric positive-definite matrix.

Then, the senders choice of distribution is characterized by $(x_\delta, \Sigma_\delta)$. In this case, his data x are sampled with respect to the distribution $\mu_\delta(x_\delta, \Sigma_\delta)$, such that:

$$\mu_\delta(x_\delta, \Sigma_\delta) = x_\delta + \Sigma_\delta^{\frac{1}{2}} \epsilon.$$

I place the following structure on data sampling. $x_\delta \in X^n$. Additionally,

1. $\sigma_i := \sqrt{\Sigma_{\delta,ii}} \in [\underline{\sigma}, \overline{\sigma}]$ and
2. $\rho_{ij} := \frac{\Sigma_{\delta,ij}}{\sigma_i \sigma_j} \in [-\overline{\rho}, \overline{\rho}]$

for $\overline{\rho} \in (0, 1)$ arbitrarily close to 1. This assumption is a multidimensional generalization of the idea that the "variance" of a variable x_i must be between $\underline{\sigma}^2$ and $\overline{\sigma}^2$. However, the key addition with n choice variables is that the sender can choose the degree of *correlation* between different variables when he samples data. I bound the correlation away from -1 and 1 to ensure the well-definedness of a solution.

Learning with Multidimensional Choices The following result generalizes receiver learning in multiple variables. As before, ϕ denotes the set of additional models R considers, given a model f_S and sampling strategies μ_δ and μ_a.

Proposition 2. *Suppose $f_S = \beta_0 + \beta_1' x - x' \delta x$ best-fits the data and $\mu_\delta = \mu_\delta(x_\delta, \Sigma_\delta)$. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + \mathrm{tr}\big((\gamma - \delta)\Sigma_\delta\big) - (x - x_\delta)'(\gamma - \delta)(x - x_\delta).$$

The proposition reads almost identically to Proposition 1, and the intuition is similar as well. However, in place of the original scalar "variance" term, we instead have $\mathrm{tr}\big((\gamma - \delta)\Sigma_\delta\big)$. To see the impact that this has on the set of models the receiver will entertain, consider the following

example with $x \in \mathbb{R}^2$.

$$\gamma = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \qquad \delta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \Sigma_\delta = \overline{\sigma}^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$$

Here, the receiver's model allows for a "cross effect" between x_1 and x_2, while the sender's model neglects it. In turn, we thus have:

$$\mathrm{tr}\big((\gamma - \delta)\Sigma_\delta\big) = \overline{\sigma}^2 \mathrm{tr}\left(\begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}\begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right) = \overline{\sigma}^2 \rho.$$

This trace term represents, as before, the *gap* between the sender's original model and the receiver's model at x_δ. The sender can *increase* this gap if he makes ρ as large as possible.

More formally, the sender in this case provides models that suffer from *neglecting interactions*. In our running example, the consultant whom the policymaker hires may effectively evaluate the effects of both a subsidy and tax *separately*, but not take into account the *interaction* of their effects.

The receiver is aware of the sender's misspecification, and she entertains additional models. However, the sender can give the receiver a more optimistic view of these models if he artificially creates a high degree of correlation between x_1 and x_2 when sampling data. In our example, if the consultant cannot figure out how to model these interactive effects, he compensates by skewing his data collection towards settings where both a carbon tax and a clean energy subsidy are active at the same time, rather than towards settings that cleanly isolate the effect of one or the other.

Results with Multidimensional Choices As before, given $f_S = \beta_0 + \beta^T x - x^T \delta x$, we denote $\Delta(f_S)$ as:

$$\Delta(f_S) = \mathbb{E}^R_\gamma[\gamma | f_S, \theta \neq a] - \delta.$$

The intuitive mapping from the one-dimensional problem to the multidimensional one can be seen by looking at the diagonal entries of $\Delta(f_S)$. If the ith diagonal entry is *positive*, this means that S tends to *neglect tradeoffs* with respect to variable x_i. If it is negative, he tends to *exaggerate* tradeoffs. Thus, we say that a sender estimates a **naive model in** x_i if $\mathbb{E}^\delta_{f_S}[\Delta_{ii}(f_S)] > 0$ and a **cautious model in** x_i if $\mathbb{E}^\delta_{f_S}[\Delta_{ii}(f_S)] < 0$.

However, what happens on the *non-diagonal* entries of Δ is also of interest. Suppose $\Delta_{ij}(\delta) + \Delta_{ji}(\delta) > 0$. This means that the receiver expects the coefficients on the interactions of x_i and x_j to be *larger* than what the sender provides. We say that this sender **neglects interactions between** x_i **and** x_j. If $\Delta_{ij}(\delta) + \Delta_{ji}(\delta) < 0$, the sender **exaggerates interactions between** x_i **and** x_j. With these new definitions in hand, we can state the key multidimensional result.

Theorem 2. *Senders who neglect interactions between x_i and x_j create collinearity between x_i and x_j. Senders who exaggerate interactions create anticollinearity between x_i and x_j. I.e.,*

- *if* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ij}(f_S) + \Delta_{ji}(f_S)] > 0$, *then* $\rho^*_{ij} = \overline{\rho}$.
- *If* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ij}(f_S) + \Delta_{ji}(f_S)] < 0$, *then* $\rho^*_{ij} = -\overline{\rho}$.

Moreover, the sender either always positions himself as a specialist in variable i ($\sigma^*_i = \underline{\sigma}$) *or generalist in variable* i ($\sigma^*_i = \underline{\sigma}$). *In particular, if* γ *is diagonal for all* $\gamma \in \Gamma$*:*

- *if* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ii}(f_S)] > 0$, *then* $\sigma_{i,\delta} = \overline{\sigma}$.
- *If* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ii}(f_S)] < 0$, *then* $\sigma_{i,\delta} = \underline{\sigma}$.

First, Theorem 2 replicates the intuitions of the univariate result: the simple sender always positions as a generalist or specialist in variable i, as opposed to pursuing an intermediate amount of sampling. In the case where there are no interaction effects (i.e. γ is always diagonal), then we obtain exactly the same result as before.

However, Theorem 2 also shows what the the sender's dataset *maximizes collinearity* between variables. If he *neglects* interactions between variables x_i and x_j, he samples data so that they are as *correlated as possible*. If he *exaggerate* interactions, he samples data so that they are *anticorrelated as possible.*

Concretely, suppose that our policymaker believes that the interaction of a carbon tax and clean energy subsidy may have particularly deleterious welfare effects at very high values, and that this is totally neglected by the consultant. The theorem states that this sender will sample data to favor settings where both a tax and subsidy are active simultaneously, rather than separately.

Intuitively, recall that any additional models the receiver entertains all have the same sample average value of y. Because the tax and subsidy are highly correlated in that data, this sample average is depressed in large part by the (negative) interaction that the sender's model neglects. To rationalize the same sample average, the policymaker's additional models must hence assign larger separate effects to the tax and subsidy than the sender's model. But since the tax and subsidy are also choice variables, the receiver need not correlate them in her eventual choice, increasing her maximal expected utility. While this paper does not explicitly deal with finite samples, in practice, the introduction of this collinearity — as well as specialist sampling — can create large standard errors and biased parameter estimates.

In this sense, when choices are multivariate, simple senders have incentives to sample data in an *unbalanced way*: rather than ensuring that the marginal effects of different variables are estimated independently of each other, there may be an arbitrary degree of collinearity between variables in the datasets that senders produce. Coupled with the previous results where senders collect data about irrelevant choices — either distant data with excessive variance or local data about a fringe receiver choice — high collinearity between variables only worsens the receiver's ability to learn.

4 Additional Results: Information and Forecasting

This section provides additional results from the baseline, univariate model. First, I characterize the *value of information* for the receiver for contracting a sender's services. I show that although this expression does not depend on whether a sender positions as a generalist or specialist, it potentially *does* depend on *where* the sender sampled data.

Second, I augment the receiver's baseline preferences with *forecast error*. As a result, it captures settings where the receiver cares not just about maximizing y but also about predicting y accurately. I show that the receiver's choices are more accurate when she receives information from a specialist than from a generalist. Consequently, the receiver may be better off hiring a cautious sender than a naive sender, although I also show that neither is first best.

Finally, Appendix B.5 studies the model with non-quadratic utility and shows that the central intuitions on sender sampling generalize. In particular, under a regularity condition, senders who estimate naive models always prefer to sample data as generalists rather than specialists, and senders who estimate cautious models prefer the opposite. Under an additional symmetry assumption, I am able to obtain the same result as Theorem 1. [12]

4.1 Receiver's Value of Information

Characterizing the value of information for the receiver from hiring a sender allows us to determine the receiver's *willingness to pay* to hire the sender, prior to data sampling and model provision. The value of information from hiring a type δ sender is given by:

$$V(\delta) := \mathbb{E}^R_{f_S|\delta}\Big[\max_x \mathbb{E}^R_{f_R|f_S}[f_R(x)|f_S(x)]\Big] - \max_x \mathbb{E}^R_{f_R}[f_R(x)] \tag{3}$$

In the absence of the sender, R would solely rely on her prior to choose x^*. In particular, she would choose $x = \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}$, which we assume is $\in [0, \overline{x}]$. This allows me to characterize the *value of information* for the receiver as follows.

Proposition 3. *Fix δ. The value of information $V(\delta)$ for the receiver from hiring a type δ sender is given by*

$$V(\delta) = \mathbb{E}^R_{f_R}\left[\mathbb{E}^R_\gamma[\gamma|f_R'(x_\delta)]\left(x^* - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right],$$

where

$$x^* = \max\left\{0, \min\left\{\overline{x}, x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f_R'(x_\delta)]}\right\}\right\}$$

[12] While the proofs of these results are significantly more involved, I also walk through techniques that are utilized in the Appendix to make the problem more tractable.

The expression for $V(\delta)$ is familiar from models of information acquisition with quadratic utility. Indeed, if the receiver perfectly learns γ, $V(\delta)$ collapses to the difference between $\mathbb{E}\left[\frac{\alpha_1^2}{\gamma}\right]$ and $\frac{\mathbb{E}[\alpha_1]^2}{\mathbb{E}[\gamma]}$.[13] However, in our setting, learning may be distorted if the sender's model is simple, when the receiver learns f_R up to its first derivative at x_δ and a constant.

Crucially, we note that $V(\delta)$ does not depend on σ, but does depend on x_δ. The intuition for the former is that different sampling strategies cause upwards and downwards shifts in the set of potential models the receiver entertains, which have no bearing on the *choice* the receiver eventually makes.[14]

The variation with respect to x_δ is of interest, although finding the optimal value of x_δ proves highly intractable. The following proposition characterizes how $V(\delta)$ varies with x_δ.

Proposition 4. *Writing the value of information $V(\delta)$ as a function of the parameter x_δ, $V(\delta, x_\delta)$, we have*

$$V_{x_\delta}(\delta, x_\delta) = -\mathbb{E}^R_{f_R}\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]\right)\left(\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}\right)^2\right].$$

Moreover if p_1^R is absolutely continuous as defined on all of $\mathbb{R}$ and $\lim_{|\alpha_1|\to\infty} \alpha_1^2 p_1^R(\alpha_1) = 0$, $\arg\max_{x_\delta} V(\delta, x_\delta) > 0$.

Intuitively, x_δ affects the receiver's ability to learn about γ through a *signal extraction problem*. That is, the receiver learns the value of $f'_R(x_\delta) = \alpha_1 - 2\gamma x_\delta$. If $x_\delta = 0$, then the receiver fully learns α_1 and places the same relative weights on different γ values as in her prior. However, if $x_\delta \neq 0$, she learns a weighted combination of α_1 and γ, with the size of the weight determined by x_δ. The nonlinearities introduced by this signal extraction problem make finding an optimal x_δ highly intractable.

4.2 Forecast Accuracy

So far, I have assumed only that the receiver would like to maximize $\mathbb{E}[y|x]$. However, in many settings, the receiver may also directly (or indirectly) value *forecast accuracy*. For example, our policymaker may not just value picking the optimal carbon tax, but also being able to *accurately forecast* the effects of her policy choice. Being able to accurately predict the effects of her policy may allow her to better defend her choice to the press and bureaucracy, as well as to be more confident in her decision.

To illustrate the impact of model simplicity and sampling on forecast-minded receivers, I consider a receiver whose expected utility is given by

$$(1-\psi)\mathbb{E}^R[y|x] - \psi \mathrm{Var}^R(y|x) \tag{4}$$

[13] With the interval types of Appendix B, R additionally learns whether γ lies below, within, or above the sender's interval; Appendix B.4 states the corresponding version of these results.

[14] Generalist and specialist sampling will, however, affect the *variance* of the receiver's outcome, which I address in the next subsection.

for $\psi \in (0,1)$. Intuitively, these preferences can also be thought of as follows. In a first period, the receiver forms an expectation of y. In a second period, given this expectation, the utility she reaps is based on her ability to forecast. If $\psi = 0$, she does not value the future or forecasting, and only cares about her expectation of y. As $\psi \to 1$, she places a greater and greater emphasis on forecasting.

Exogenous Forecast Error Before studying what happens when the receiver chooses x to maximize (4), I first characterize how the variance of y behaves with respect to σ_δ when the receiver simply maximizes the expected value off y. The following proposition shows that, fixing x_δ, the variance of outcomes is minimized at an *intermediate* value of σ_δ. That is, neither specialist nor generalist sampling minimizes forecast error for the receiver. [15]

Proposition 5. *Fix a sender type δ and let $x^*(f_S)$ maximize $\mathbb{E}_{f_R|f_S,\mu_\delta}[y|x]$. Then, there exists $\sigma_\delta^* \in (\underline{\sigma}, \overline{\sigma})$ that maximizes*

$$W(\sigma_\delta) := \mathbb{E}^R_{f_S|\delta}\Big[\mathrm{Var}^R_{f_R|f_S}\big(y|x^*(f_S), x_\delta, \sigma_\delta\big)\Big].$$

with respect to σ_δ. Moreover, W is strictly decreasing in σ for $\sigma < \sigma_\delta^*$ and increasing for $\sigma > \sigma_\delta^*$.

The intuition for the result is as follows. Writing x^* as the value of x that maximizes the expectated value of y, recall our previous formulation that

$$\mathbb{E}[f_R(x^*)|f_S] = f_S(x^*) + \Delta(f_S)\big(\sigma^2 - (x^*(f_S) - x_\delta)^2\big)$$

The receiver's forecast error is driven by the latter set of terms multiplied by $\Delta(f_S)$. In particular, the variance of outcomes can be expressed as the difference between the degree of sampling σ^2 and the squared distance between x^* and x_δ (the optimal choice and the point around which data are sampled). When σ^2 is small, the dispersion is driven by $(x^* - x_\delta)^2$, which is always nonzero in expectation, meaning that the variance can be reduced by increasing σ^2. Intuitively, because there is uncertainty about what the ex-ante optimal choice x^* will be, an intermediate degree of sampling allows the receiver to "hedge" and gather information in a slightly larger region where she expects the choice to be, rather than precisely at a single point. When σ^2 is very large, however, there is a great deal of vertical dispersion in the potential values of y that may materialize at x^*, and that dispersion can be reduced by decreasing σ^2.

Thus, the *second-best* degree of sampling σ_δ^* that the receiver would choose to minimize outcome variance is necessarily intermediate: that is, neither a sender who samples as a generalist nor a specialist minimizes the variance of the receiver's outcomes. However, we note that as $\overline{\sigma}$ grows arbitrarily large, the sender *will* be better off under specialist sampling than generalist sampling.

[15] As before, comparative statics with respect to x_δ are not straightforward, and so we omit them throughout.

Endogenous Forecast Error We now study the setting where R jointly trades off her expected gain in y with forecast error. Conditional on f_S and fixing x_δ, her expected value is

$$V^F(f_S|\sigma_\delta) = \max_x (1-\psi)\mathbb{E}_{f_R|f_S,\sigma_\delta}[y|x] - \psi \text{Var}_{f_R|f_S,\sigma_\delta}[y|x].$$

We then study the sender's behavior when the receiver has these preferences.

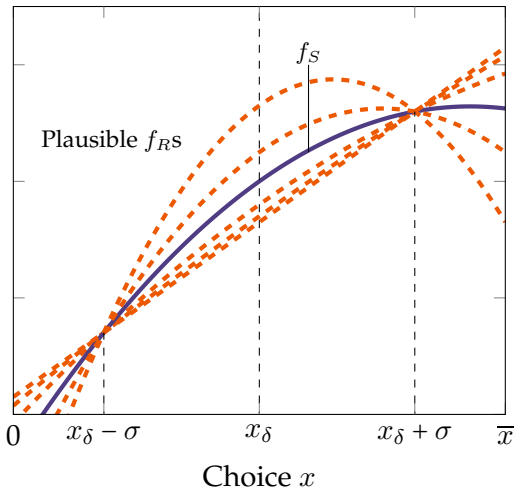
Proposition 6. *Suppose R chooses $x^*(f_S)$ to maximize (4). Then:*

1. *A naive sender chooses $\sigma_\delta > \sigma_\delta^*$ and a cautious sender chooses $\sigma_\delta < \sigma_\delta^*$*
2. *As $\psi \to 1$, $x^*(f_S) \to x_\delta \pm \sigma$.*

The first part of the result generalizes the intuition of Theorem 1. Namely, senders who estimate naive models sample data with a *larger spread* than what the receiver would choose, and those who estimate cautious models sample with *less spread*. The intuition is that the sender accounts for the fact that the receiver will shift her choices to take into account forecast error. For very high values of σ^2, a naive sender trades off between increasing the receiver's expectation of y through generalist sampling and increasing the variance of outcomes she faces. The opposite logic holds for senders who estimate cautious models.

The second part of the result shows that if the receiver increasingly only values forecast error, her choice approaches $x_\delta \pm \sigma_\delta$. This is illustrated in Figure 6. Thus, R trades off between a choice $x = x^\dagger$ that maximizes her expected utility (at an interior point) and minimizing the variance of outcomes, which happens at $x_\delta \pm \sigma$. When σ is very large, R is forced to choose arbitrarily extreme points $x_\delta \pm \sigma$ to minimize variance. When σ is very small, R chooses points close to x_δ.

Figure 6: Intersection of receiver's models with sender's models



5 Gains from Simplification

This section returns to the univariate quadratic model to investigate gains from model simplicity. So far, I have taken model simplicity as exogenous: simple sender types present models that are artificially naive or cautious. This section characterizes the degree to which the accurate sender benefits from the presence of other, simpler senders, and shows that this gain is maximized when simple senders are *polarized* into highly naive and cautious groups.

In particular, I ask what kind of *market* for models a receiver may endogenously face: i.e., how likely she may be to receive an *accurate* or *simple* model, whether simple models tend to be naive or cautious, and how frequent each model type is in the population of senders she faces.

In particular, prior to being hired by the receiver, I endogeneously allow the accurate sender a to choose the *distribution* of simple senders. The interpretation is that the accurate sender is able to hire additional senders alongside him who may provide simple models. Conditional on contracting the services of his organization, a receiver may receive information from either the accurate sender or one of these simple senders. For example, the accurate sender could be thought of as:

- a consultant or portfolio manager who can employ other consultants/managers whose recommendations may take the form of simple rules of thumb;
- a politician who can recruit co-partisans to propagate simple talking points among voters;
- an AI company that can deploy simpler or more complex LLMs through its chatbot.

The key is that he is able to *shape* the receiver's beliefs about whether she is facing a simple or accurate model.

I show that the accurate sender would organize the field of senders as follows. First, he would either provide his own services, or those of "simple" senders who provide models that are calibrated to arbitrary parameters. Formally, the type space of simple senders is given by Γ, and never includes any "interval types" as studied in Section B. Second, the *distribution* of these types is characterized by *polarization* in how senders consider tradeoffs. Formally, for each $\delta \in [0, \overline{\gamma}]$, the conditional probability of facing a type δ sender is given by a density $t(\delta)$. This density $t(\delta)$ takes the accurate sender's prior over δ and pushes mass away from intermediate values of δ towards the edges of $[0, \overline{\gamma}]$. I.e., simple senders either estimate models with low δ values (close to 0) or high δ values (close to $\overline{\gamma}$), and never estimate models with intermediate δ values. Thus, the set of simple sender types is fully composed of senders who are (arbitrarily) naive or cautious in the models they estimate. As the probability of seeing the accurate sender's model approaches 1, the type space of simple senders converges to a one- or two-point distribution concentrated on either $\delta = 0$, $\delta = \overline{\gamma}$, or both points. These respectively correspond to senders who only provide "linear regression" models or "worst-case scenario" models, as I studied in Section 3.

5.1 Motivation

To motivate the role of the accurate sender further, we return to the problem of the consultant and the policymaker. Imagine an "accurate" consultant employed by a consulting organization, who can think about tradeoffs from carbon taxation in *exactly* the way the policymaker would like. However, he can also choose to hire additional colleagues, who might think about tradeoffs in a simpler way than both himself and the policymaker. He can choose how many of these colleagues to hire, as well as how simply they might think about problems.

The consultant thus hires other consultants. The policymaker, in turn, is aware of the composition of the consultant's organization. When she contracts the organization's services, she is randomly assigned either the "accurate" consultant or one of the "simple" consultants. Relatedly, she may contract a team composed of both the accurate consultant as well as other simple consultants, but not know how much oversight the accurate consultant exercised. Based on the model she is ultimately provided, she in turn attempts to *infer* the sender's type. Thus, our problem boils down to: will the accurate consultant hire "simple" consultants to join him, what kind of simplicity will they adopt, and how many of those consultants will he hire?

The key interpretation of the design problem is thus that an accurate sender can endogenously shape the set of senders that the receiver considers, such as by actively hiring additional senders who model tradeoffs differently. Beyond hiring, we also highlight additional interpretations based on our motivating examples.

1. **Portfolio management.** Suppose our sender is a portfolio manager who can estimate a highly accurate model of asset returns. The manager may help train other managers or deploy "robo-advisors" that use simple rules of thumb.

2. **Political narratives.** Suppose our sender is a politician who can accurately provide a "narrative model" of a policy's effects on voter welfare. However, the same politician can also propagate "simple narratives" of a policy's effects through online media accounts or by aligning himself with other politicians who provide simple narratives.

3. **AI recommendations.** Suppose our sender is an AI platform that uses a sophisticated LLM to collect data and estimate a robust model that provides users accurate recommendations. The platform can also deploy simpler LLMs that apply a "fixed" way of thinking about a user's problem, and can determine how often users are shown each model under the hood.

The key mechanism through which this design problem will operate is through manipulating the receiver's *beliefs* about whether the model she sees came from an accurate or simple sender.

5.2 Constructing the Market for Senders

Design Problem Setup Formally, consider the following problem for the accurate sender type a sender. Conditional on a model f_S, with probability p_a, R believes she has received a model from

a. With probability $1 - p_a$, she believes the model came from a type space Θ of simple senders. a can design the type space Θ. That is, his problem is to choose a set $\Theta \subseteq [0, \overline{\gamma}]$ of sender types.[16] We write $P^a_{f_S}$ for his prior over models f_S. His objective is:

$$\max_{\Theta, T} p_a \int_{f_S \in F_R} \max_x \mathbb{E}_{q|f_S,T,\Theta}[f_R(x)] dP^a_{f_S}(f_S).$$

As before, the type a sender wants to make the receiver optimistic about the expected value of her choice. We note that the impact of Θ is through the receiver's *posterior*. I take p_a as fixed; this could represent the type a sender's capacity to hire others (higher p_a means he has less latitude to hire), the exogenous choice of a platform/organization, or a market belief.[17]

The order of events is as follows. a first chooses a type space Θ. Then, a chooses a distribution over types T. The receiver contracts the services of a sender, and receives a model from a with probability p_a or from the type space Θ with probability $1 - p_a$. Conditional on the latter event, a type δ sender is then drawn according to the distribution T. Finally, the deployed sender strategically samples data as before, supplies the model of best fit, and provides it to the receiver.

Optimal Type Space We note that because the type a sender's models are not misspecified, his data sampling strategy does not affect the alternative models R considers. However, a "simple" sender with type δ will always strategically sample.[18] Thus, the assumption that the receiver cannot directly see the sender's data is key.

In turn, the optimal characterization of the type space is as follows. Appendix B.3 shows that the first part of this result continues to hold when a can also create types that restrict curvature to an interval of values: he never does so.

Theorem 3. *Suppose P^a_δ is continuous. Then, the optimal type space is $\Theta = [0, \overline{\gamma}] = \Gamma$. Moreover, there exists a function $\pi(\delta)$ such that T is given by a continuous distribution t with*

$$t(\delta) = p^a_\delta(\delta)\pi(\delta). \tag{T^*}$$

In particular, there exist thresholds $0 \leq \underline{\delta}_0 < \mathbb{E}^R_\gamma[\gamma] < \overline{\delta}_0 \leq \overline{\gamma}$ such that $\pi(\delta)$ is strictly decreasing below $\underline{\delta}_0$, strictly increasing above $\overline{\delta}_0$, and equal to 0 on $[\underline{\delta}_0, \overline{\delta}_0]$.

I first interpret the result before describing its mathematical intuition. Theorem 3 says that the optimal type *space* is given by every curvature value. For each $\delta \in [0, \overline{\gamma}]$ the accurate sender/designer creates a sender of type δ. Recalling a prior definition, this sender has the interpretation that he *calibrates* the value of δ to a fixed (but arbitrary) value in $[0, \overline{\gamma}]$. This means that conditional on

[16] Appendix B.3 allows a to also choose types that are intervals of curvature values, and shows that he would never do so.

[17] p_a could also endogenously be chosen by the type a sender himself to trade off the probability with which his services will be contracted with receivers' perceptions of him.

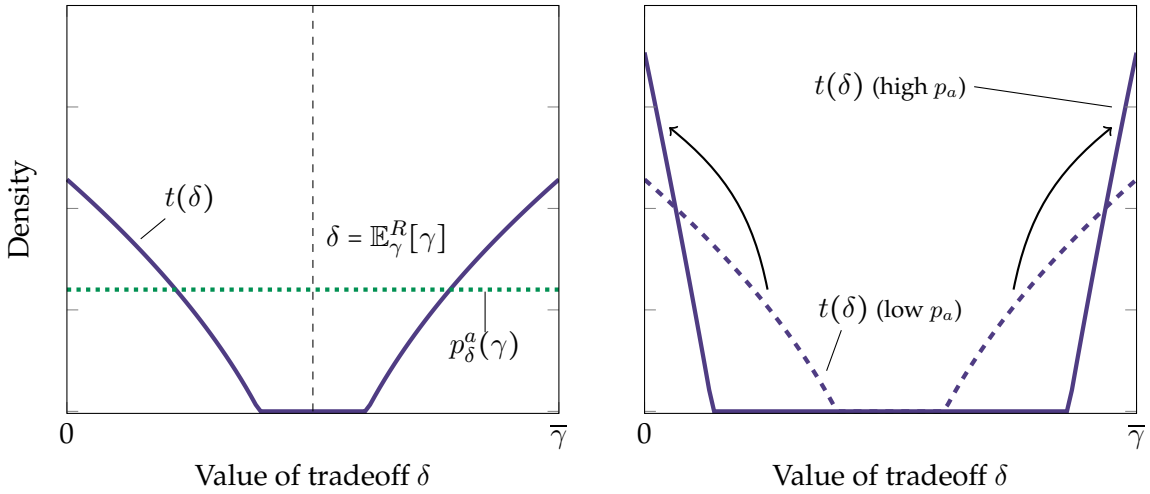
[18] If the sender is such that the expected value of $\Delta(f_S) = 0$, we assume that sender samples at 0.

seeing a model f_S, the receiver believes it is either the true model or was calibrated to an *arbitrary* value.

The second part of the theorem characterizes the *shape* of the distribution of δ values the type a sender utilizes. The solution is that the type a sender takes his prior over $[0, \overline{\gamma}]$ and *warps it* in a way that pushes more mass to the *edges* of the interval. In this sense, the type a sender *polarizes* simple senders towards models calibrated to be *arbitrarily* naive (with δ close to 0) or cautious (with δ close to $\overline{\gamma}$). In particular, $t(\mathbb{E}^R_\gamma[\gamma]) = 0$, i.e. t puts 0 mass on types with δ equal to the receiver's prior expectation of γ.

This idea is shown on the left panel of Figure 7 below. The dotted green line shows the *designer*'s prior distribution over δ, which is given by a continuous uniform distribution on $[0, \overline{\gamma}]$. The solid purple line shows that the types he creates are concentrated towards the edges at $\delta = 0$ (i.e. linear models) and $\delta = \overline{\gamma}$ (i.e. worst-case models).

Figure 7: Prior distribution $p^a_\delta(\delta)$ of γ vs. optimal distribution of sender types $t(\delta)$



(a) Optimal distribution of sender types $t(\delta)$ vs. designer's prior over tradeoffs $p^a_\delta(\delta)$

(b) Increased polarization of $t(\delta)$ with increase in likelihood of seeing true model p_a

The first part of the result relies on the intuitions of Proposition **??**. Consider a model of the form $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$. For each δ, we introduce a sender of type δ. This sender is simple in the sense that he provides models with an arbitrarily fixed value of δ. Conditional on seeing $f_S(x)$, R would believe that it was either a true model from the accurate sender or the same model provided by the simple type δ sender. In the former case, the receiver would believe f_S is the true model. In the latter case, because the simple sender sampled strategically, R's maximal expected utility from considering alternative models would be higher than from just considering f_S. the type a sender thus always gains from creating a simple sender for each $\delta \in \Gamma$.

The intuition for the second part of the result is as follows. The accurate sender's problem is to choose a distribution over types which, in this case, boils down to just being a distribution over the set of types δ. In particular, he designates an amount of probability density to allocate to each

$\delta \in [0, \overline{\gamma}]$, subject to the constraint that these densities must integrate out to 1.

Suppose $\delta = \mathbb{E}^R_\gamma[\gamma]$, which is the *receiver*'s ex-ante expectation of γ. Looking back to Proposition 1, simple senders gain neither from generalist nor specialist sampling in this case. Relatedly, this means that the type a sender has no incentive to place mass on δ values near $\mathbb{E}^R_\gamma[\gamma]$. On the other hand, as δ gets farther and farther away from $\mathbb{E}^R_\gamma[\gamma]$, simple senders provide a larger and larger gain from simple models. In fact, the gains to the sender are conditionally maximal if $\delta = 0$ or $\delta = \overline{\gamma}$. This is what drives the "polarization" of the type space.

Variation with p_a We note that as p_a increases, the receiver is more likely to think that she saw the accurate sender/designer's model, and that as p_a decreases, this is less likely. In the example with the consultant and policymaker, our "designer" may be required to provide his services p_a of the time, but has latitude to choose the senders who the receiver contracts $1 - p_a$ of the time. As p_a increases, this latitude decreases, and the receiver is conditionally more and more sure that she received the true model. The following proposition shows that simple sender "polarization" is increasing in p_a, in the sense that as p_a increases, $t(\delta)$ places more and more mass on extreme types with very high or low δ values, and less (or none) on intermediate types.

Proposition 7. *The polarization of the type space is increasing in p_a. That is, $\underline{\delta}_0$ is decreasing in p_a and $\overline{\delta}_0$ is increasing in p_a. As $p_a \to 1$, $t(\delta)$ converges to a distribution that either places all mass only on $\delta = 0$, $\delta = \overline{\gamma}$, or both. As $p_a \to 0$, $t(\delta)$ converges to a distribution that is strictly decreasing on $[0, \mathbb{E}^R_\gamma[\gamma])$, equal to 0 at $\mathbb{E}^R_\gamma[\gamma]$, and strictly increasing on $(\mathbb{E}^R_\gamma[\gamma], \overline{\gamma}]$.*

The proposition is visualized above in the right panel of Figure 7. The dashed purple line graphs $t(\delta)$ for a lower value of p_a. The solid line graphs $t(\delta)$ for a higher value of p_a, showcasing the "increased polarization" generated by an increase in p_a.

The intuition is as follows. the type a sender's gain from being mistaken for a simple sender is high at $\delta = 0$, decreases to 0 at $\delta = \mathbb{E}^R_\gamma[\gamma]$, and then rises again until $\delta = \overline{\gamma}$. That is, he conditionally gains the most when δ is very low or very high. When p_a is large, the ex-ante probability with which he will be seen as a simple sender is low, and he is thus best off concentrating simple senders close to $\delta = 0$ and $\delta = \overline{\gamma}$, where he expects the gain to be largest. However, when p_a is small, he is willing to distribute simple senders more evenly throughout the population although, in either case, he concentrates few senders near $\delta = \mathbb{E}^R_\gamma[\gamma]$.

I highlight two additional features of this result. First, polarization survives even as $p_a \to 0$ (so the receiver is less likely to think she sees the true model). Second, as $p_a \to 1$, the optimal distribution either converges to only deploying senders with type 0 or type $\overline{\gamma}$ and nothing in between. That is, the model returns precisely to the two-type example introduced in Section 2: in the limit, the only simple senders are those who *only provide* linear regression models or *only provide* "worst-case" models.

6 Conclusion

This paper studies how strategic data collection and model simplicity shape communication with models and data. I study a theoretical model where a receiver R contracts a sender S to help her make a decision. The sender S first collects data and then estimates a model. However, R does not necessarily know what kind of data S estimated the model on. S's model may additionally be *simplified* in that it is *naive* and neglects tradeoffs R cares about — i.e. always assumes models have lower curvature —- or is *cautious* and exaggerates the extent of tradeoffs — i.e. always assumes models with high curvature. S's goal is to make R as optimistic as possible. The setting I study has a wide variety of applications to the economics of organizations, portfolio management, political economy, delegation in policymaking, and the use of AI-based decisionmaking.

I first characterize how R learns based on the model that S provides her. I show that R may sometimes believe that S provided an accurate model of her decisions, but that more often than not, R must entertain additional models. If S estimated his model on a small set of choices as a "specialist," I show that the additional models R entertains are all tangent to S's original model at a point. If S estimated his model on a large range of choices as a "generalist," I show that the additional models R entertains are all secants through S's original model.

I then show a close correspondence between what kinds of data S collects and his model *simplicity*. If S estimates naive models that *neglect* tradeoffs, then he collects data as a generalist across an arbitrarily wide set of a receiver's choices. If S estimates cautious models that *exaggerate* tradeoffs, then he collects data as a specialist locally about a single extreme receiver choice. In both cases, the receiver considers not only the sender's original model, but also models that deliver strictly higher expected utility, creating gains from the coupling of model simplicity and strategic data sampling. Thus, there is a close correspondence between naive models and positioning oneself as a "generalist"; and between cautious models and positioning oneself as a "specialist." When choices are multivariate, I show that the sender samples data in a way that maximizes collinearity between variables, and makes them as correlated (or anticorrelated) as possible.

Taken together, these points suggest that the quality of *data* may often be *low* when contracting external senders to present model-based recommendations. Moreover, this low quality of data emerges *precisely* as a result of the fact that senders estimate simple models. In particular, the data underlying simple models will often be only loosely related to the choices receivers eventually make. Generalist senders sample data as widely as possible across the choice space and, in particular, place as much mass as possible at its edges. Specialist data are precise, but only describe the value of outcomes at one edge of the choice space. In both cases, the data embed information about choices the receiver is unlikely to end up making. This means that clients should be wary when interpreting the breadth of a consultant's or portfolio manager's data, particularly if they believe that the consultant or portfolio manager did not think about the potential costs of actions as much as they would like. A politician who presents voters a naive "narrative of everything" may only ever be accurate in some cases. On the flipside, even when presented with a "worst-case scenario," clients should themselves be cautious about whether that data idiosyncratically speaks

to an extreme action — such as very high or very low taxes or fringe investment strategies.

The results on multivariate choices show further that the data senders collect may be particularly unbalanced and highly collinear. Although the paper abstracts from finite samples, high collinearity degrades the precision of parameter estimates and the balance of data across covariates, and thus the accuracy of decisions. In our example, a consultant who cannot model how a carbon tax and a clean energy subsidy interact draws evidence from settings where both were adopted together; this leaves the policymaker with little information about what each policy does on its own. More broadly, policymakers may be provided evidence about the effects of multiple reforms operating at once. A firm that hires a strategy consultancy may receive a model based on situations where both advertising campaigns and price promotions were run at the same time. This collinearity makes it difficult to separate the levers that these different choices control.

The paper further shows that the degree of sampling that minimizes forecast error for the sender is an *intermediate* degree of data collection between pure generalism and specialization. Hiring either a specialist or generalist sender is likely to result in imprecise learning, even conditional on a sender's misspecification. This illustrates a tradeoff between optimism and accuracy: the models that make receivers most optimistic do not always lead to the best choices. A CEO who hires the consultant who makes her potential decisions look best may also struggle to pin down the effects of her choices, affecting her ability to justify that decision to a board of directors.

I show that senders who provide the true model can generally gain from the existence of simple senders. That is, if a sender provides a true, accurate model of a receiver's choices, he always gains if the receiver thinks his model was (artificially) naive or cautious. The intuition is that, in the latter cases, the receiver entertains not only a sender's original model but also additional models that lie above it. Thus, the sender who provides an accurate model can make the receiver have a more optimistic belief about her maximal expected utility than the true model. Accurate senders may therefore try to masquerade as simple ones: clients of a large consulting firm may not know whether a senior partner or junior analyst built their model, investors may be unable to tell whether an experienced manager or a robo-advisor suggested a rule, and users of an AI platform may not know whether a recommendation came from a small or a large model.

Finally, I use these results on model simplicity to characterize the market for senders. I consider the problem of an accurate sender who estimates an accurate model of a receiver's tradeoffs. However, he can also surround himself with "simple" senders who estimate naive or cautious models. First, I show (in Appendix B) that whenever the accurate sender wants to hire simpler senders, he always hires senders who *calibrate* their models to arbitrary parameters, and thus never truly estimate the value of tradeoffs on data. In the real world, these correspond to senders who calibrate their models to pre-defined parameters: economic consultants may calibrate models to elasticities or labor shares from the literature rather than organically estimating them in their setting; robo-advisors may apply fixed investing rules of thumb across clients; and politicians may weave narratives in which a policy's costs are fixed in advance. The paper suggests that calibration of this kind need not reflect a lack of data or expertise, but may instead emerge as a feature of

a market for senders that advantages those able to provide the most accurate models.

Second, I show simple senders polarize into two distinct groups: those who estimate (arbitrarily) naive models that are close to "linear regression" models, and those who estimate (arbitrarily) cautious models that are close to "worst-case" models. Thus, the market of senders is composed of three groups: some senders who always provide accurate models, some who provide highly naive models, and some who provide highly cautious models. In the limit, a receiver sees only linear regressions or worst-case models. These simple models tend to recommend either very extreme or very conservative actions: naive models suggest making choices as extreme as possible, while cautious models may imply sticking to the status quo. While this form of polarization is common among political narratives, the paper suggests that it may occur more generally when an external party is hired to provide a model or *narrative* of the effects of different choices. A market for models may be saturated with models that recommend very extreme or conservative choices.

The combination of simple models, strategic data sampling, and polarization all represent key frictions that may emerge when decisionmakers rely on external expertise and modeling to shape their choices. Future work will investigate these frictions further in the settings of price-based markets and the polarization of political platforms, as well as through empirical work that investigates the use of AI-based decisionmaking in firms or the role of consultants' narratives in shaping organizational decisionmaking.

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A Proofs of Results

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ best-fits the data and the simple sender's sampling strategy μ_δ has mean and variance x_δ and σ_δ^2. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + (\gamma - \delta)\big(\sigma_\delta^2 - (x - x_\delta)^2\big).$$

Proof. We rewrite the sender's model as a local expansion around x_δ: $f_S(x) = \tilde{\beta}_0 + \tilde{\beta}_1(x - x_\delta) - \delta(x - x_\delta)^2$, where $\tilde{\beta}_1 = \beta_1 - 2\delta x_\delta$. Because a sender's type is the curvature δ of his model, the value of δ uniquely pins down the sender's type.

Suppose $f_R \in \phi(f_S; \mu_\delta)$. Similarly expanding f_R around x_δ, we have:

$$f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1(x - x_\delta) - \gamma(x - x_\delta)^2.$$

Because μ_δ is symmetric, it is elliptical (in one dimension). By Stein's Lemma (Landsman and Nešlehová, 2008), equation (2) can be written as

$$\begin{aligned}\mathbb{E}_{\mu_\delta}\big[x\big(f_R(x) - f_S(x)\big)\big] &= \mathrm{Cov}_{\mu_\delta}(x, f_R(x) - f_S(x)) \\ &= \sigma^2 \mathbb{E}_{\mu_\delta}[f_R'(x) - f_S'(x)] \\ &= \sigma^2 \mathbb{E}_{\mu_\delta}[\tilde{\alpha}_1 - 2\gamma(x - x_\delta) - \tilde{\beta}_1 + 2\delta(x - x_\delta)] = 0\end{aligned}$$

Because the mean of μ_δ is x_δ, this yields the condition $\tilde{\alpha}_1 = \tilde{\beta}_1$ which is *independent* of σ^2. This means that $\alpha_1 - 2\gamma x_\delta = \beta_1 - 2\delta x_\delta$, i.e. $\alpha_1 = \beta_1 - 2(\delta - \gamma)x_\delta$. Hence, we have

$$f_R(x) = \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\delta)x - \gamma x^2,$$

with the value of γ and α_0 potentially indeterminate. However, fixing a plausible γ value, from (1) we have

$$\begin{aligned}\mathbb{E}_{\mu_\delta}[f_R(x)] = \alpha_0 + (\beta_1 - 2(\delta - \gamma)x_\delta)x_\delta - \gamma(x_\delta^2 + \sigma^2) = \beta_0 + \beta_1 x_\delta - \delta(x_\delta^2 + \sigma^2) = \mathbb{E}_{\mu_\delta}[f_S(x)] \\ \implies \alpha_0 - 2(\delta - \gamma)x_\delta^2 = \beta_0 - \delta x_\delta^2 - \delta\sigma^2 \\ \implies \alpha_0 = \beta_0 + (\gamma - \delta)(\sigma^2 - x_\delta^2).\end{aligned}$$

Putting the pieces together, we have

$$\begin{aligned}f_R(x) &= \beta_0 + (\gamma - \delta)(\sigma^2 - x_\delta^2) + (\beta_1 - 2(\delta - \gamma)x_\delta)x - \gamma x^2 \\ &= \beta_0 + \beta_1 x - \delta x^2 + \big(-(\gamma - \delta)x^2 + (\gamma - \delta)(\sigma^2 - x_\delta^2) + 2(\gamma - \delta)x_\delta\big) \\ &= \beta_0 + \beta_1 x - \delta x^2 + (\gamma - \delta)\big(\sigma^2 - (x - x_\delta)^2\big) \\ &= f_S(x) + (\gamma - \delta)\big(\sigma^2 - (x - x_\delta)^2\big).\end{aligned}$$

□

Theorem 1. *If a type δ simple sender S estimates naive models, he positions himself as a generalist ($\sigma^*_\delta = \overline{\sigma}$) and sets $x^*_\delta = \mathbb{E}^\delta_{f_S}[x^*(f_S)]$, where $x^*(f_S)$ is the sender's expectation of the receiver's choice given f_S. If he estimates cautious models, he positions himself as a specialist ($\sigma^*_\delta = \underline{\sigma}$) and sets $x^*_\delta = 0$ or $\overline{x}$.*

Proof. Following from Proposition 1, given a type δ sender's model, any $\gamma \in [0, \overline{\gamma}]$ value is possible from the receiver's perspective. This pins down the plausible set of γ values from the receiver's perspective.

Next, fix σ. The receiver chooses $x = x^*$ to maximize:

$$\begin{aligned}&\beta_0 + \beta_1 x - \delta x^2 + \big(\mathbb{E}^R_\gamma[\gamma|f_S] - \delta\big)\big(\sigma^2 - (x - x_\delta)^2\big)\\ &= \beta_0 + \beta_1 x - \delta x^2 + \Delta(f_S)\big(\sigma^2 - (x - x_\delta)^2\big)\end{aligned} \tag{5}$$

We note that because the sender behaves as if the receiver has a flat prior over α_0, her posterior places the same mass on all models that are the same "up to a constant"; hence, changing σ^2 does not change the relative mass she places on each model, fixing a plausible γ value. Thus, by the Envelope Theorem, (5) is increasing in σ^2 if $\Delta(f_S)$ is positive and decreasing if $\Delta(f_S)$ is negative. Working backwards, when the sender takes expectations over (5), he chooses $\sigma = \overline{\sigma}$ if $\mathbb{E}_{f_S}[\Delta(f_S)] \geq 0$ and $\underline{\sigma}$ otherwise.

Finally, we characterize the equilibrium value of x_δ. Note that because the sender behaves as if the receiver's prior is flat over α_0 and α_1, his expectation of the receiver's expectation of γ does not depend on x_δ. Returning to (5) and using the envelope theorem, we have that the derivative with respect to x_δ is:

$$2\Delta(f_S)(x^* - x_\delta)$$

This is (strictly) concave if $\Delta(f_S) > 0$, and achieves a maximum when $x^* = x_\delta$. If $\Delta(f_S) \leq 0$, the function is convex and achieves a maximum at a corner solution 0 or $\overline{x}$, giving the potential values for x_δ. Given S must commit to data collection beforehand and integrating backwards, this gives $x_\delta = \mathbb{E}^\delta_{f_S}[x^*]$ if he expects $\Delta(f_S) > 0$ and $x_\delta = 0$ or $\overline{x}$ otherwise.

Finally, we note that $x^*(f_S)$ solves

$$\begin{aligned}\beta_1 - 2\delta x^* - 2\Delta(f_S)(x^* - x_\delta) = 0 \implies x^* &= \frac{\beta_1 + 2\Delta(f_S)x_\delta}{2\mathbb{E}^R_\gamma[\gamma|f_S]}\\ \implies \mathbb{E}^\delta_{f_S}[x^*(f_S)] &= \mathbb{E}^\delta_1\left[\frac{\beta_1 + 2\Delta(f_S)x_\delta}{2\delta + 2\Delta(f_S)}\right].\end{aligned}$$

□

Proposition 2. *Suppose $f_S = \beta_0 + \beta_1' x - x'\delta x$ best-fits the data and $\mu_\delta = \mu_\delta(x_\delta, \Sigma_\delta)$. Then, for all $f_R \in \phi(f_S; \mu_\delta, \mu_a)$, we have:*

$$f_R(x) = f_S(x) + \operatorname{tr}\big((\gamma - \delta)\Sigma_\delta\big) - (x - x_\delta)'(\gamma - \delta)(x - x_\delta).$$

Proof. As above, we necessarily have the following moment conditions for all i:

$$\mathbb{E}_{\mu_\delta}\big[x_i\big(f_R(x) - f_S(x)\big)\big] = 0 \qquad \mathbb{E}_{\mu_\delta}\big[f_R(x) - f_S(x)\big] = 0$$

and we may or may not have the correct moment condition with respect to the entries of γ. Crucially, however, because μ_δ is elliptical by Stein's Lemma for elliptical random vectors (Landsman and Nešlehová, 2008), we can simplify the first equation to

$$\begin{aligned}
\mathbb{E}_{\mu_\delta}\big[x_i\big(f_R(x) - f_S(x)\big)\big] = 0 &\implies \mathbb{E}_{\mu_\delta}\big[\frac{\partial}{\partial x_i} f_R(x)\big] = \mathbb{E}_{\mu_\delta}\big[\frac{\partial}{\partial x_i} f_S(x)\big] \qquad \forall i \\
&\implies \alpha - (\gamma + \gamma')\mathbb{E}_{\mu_\delta}[x] = \beta - (\delta + \delta')\mathbb{E}_{\mu_\delta}[x] \\
&\implies \alpha = \beta + \big((\gamma + \gamma') - (\delta + \delta')\big)x_\delta \\
&\implies \alpha' x = \beta' x + x_\delta'\big((\gamma + \gamma') - (\delta + \delta')\big)x
\end{aligned}$$

which is invariant to Σ_δ. Next, working with the second equation, we have

$$\mathbb{E}_{\mu_\delta}\big[f_R(x)\big] = \alpha_0 + \alpha x_\delta - \big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big)'\gamma\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big) = \beta_0 + \beta x_\delta - \big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big)'\delta\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big) = \mathbb{E}_{\mu_\delta}\big[f_S(x)\big].$$

Note that

$$\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big)'\gamma\big(x_\delta + \Sigma_\delta^{\frac{1}{2}}\epsilon\big) = x_\delta'\gamma x_\delta + 2x_\delta'\gamma\Sigma_\delta^{\frac{1}{2}}\epsilon + \Sigma_\delta^{\frac{1}{2}}\epsilon'\gamma\Sigma_\delta^{\frac{1}{2}}\epsilon$$

and taking the expectation with respect to ϵ thus gives

$$x_\delta'\gamma x_\delta + \operatorname{tr}(\gamma\Sigma_\delta).$$

Utilizing the same logic with δ and plugging in the α equation gives

$$\begin{aligned}
\alpha_0 &= \beta_0 + (\beta - \alpha)x_\delta + x_\delta'(\gamma - \delta)x_\delta + \operatorname{tr}((\gamma - \delta)\Sigma_\delta) \\
&= \beta_0 - x_\delta'\big((\gamma + \gamma') - (\delta + \delta')\big)x_\delta + \operatorname{tr}((\gamma - \delta)\Sigma_\delta) \\
&= \beta_0 - x_\delta'(\gamma - \delta)x_\delta + \operatorname{tr}((\gamma - \delta)\Sigma_\delta)
\end{aligned}$$

Hence, we renormalize f_R, plug in the values of α_0, α, and reorganize the quadratic terms to get:

$$\begin{aligned}
f_R(x) &= \alpha_0 + \alpha' x - x'\gamma x \\
&= \beta_0 - x_\delta'(\gamma - \delta)x_\delta + \operatorname{tr}((\gamma - \delta)\Sigma_\delta) + \beta' x + x_\delta'\big((\gamma + \gamma') - (\delta + \delta')\big)x - x'\gamma x \\
&= \beta_0 + \beta' x + \operatorname{tr}((\gamma - \delta)\Sigma_\delta) - (x - x_\delta)'(\gamma - \delta)(x - x_\delta).
\end{aligned}$$

□

Theorem 2. *Senders who neglect interactions between x_i and x_j create collinearity between x_i and x_j. Senders who exaggerate interactions create anticollinearity between x_i and x_j. I.e.,*

- *if* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ij}(f_S) + \Delta_{ji}(f_S)] > 0$, *then* $\rho^*_{ij} = \overline{\rho}$.
- *If* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ij}(f_S) + \Delta_{ji}(f_S)] < 0$, *then* $\rho^*_{ij} = -\overline{\rho}$.

Moreover, the sender either always positions himself as a specialist in variable i ($\sigma^*_i = \underline{\sigma}$) or generalist in variable i ($\sigma^*_i = \underline{\sigma}$). In particular, if γ is diagonal for all $\gamma \in \Gamma$:

- *if* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ii}(f_S)] > 0$, *then* $\sigma_{i,\delta} = \overline{\sigma}$.
- *If* $\mathbb{E}^{\delta}_{f_S}[\Delta_{ii}(f_S)] < 0$, *then* $\sigma_{i,\delta} = \underline{\sigma}$.

Proof. The proof of this result follows is similar to that of Theorem 1. The problem for a type δ sender is given by:

$$\max_{\Sigma_\delta, x_\delta} \mathbb{E}^{\delta}_{f_S}\left[\max_{x\in[0,\overline{x}]} \beta_0 + \beta' x + \Big(1 - \Pr(\theta = a|f_S)\Big)\Big(\operatorname{tr}((\gamma-\delta)\Sigma_\delta) - (x - x_\delta)'(\gamma-\delta)(x-x_\delta)\Big)\right]$$

subject to the constraints that Σ_δ is positive definite and variances/correlations are in their bounded intervals:

$$\Big(\prod_{i=1}^{j}\sigma_i^2\Big)\det\begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1j} \\ \rho_{12} & 1 & \cdots & \rho_{2j} \\ \vdots & & \ddots & \vdots \\ \rho_{1j} & \rho_{2j} & \cdots & 1 \end{pmatrix} > 0 \quad \text{for } j = 2,\dots,n, \quad \sigma_i \in [\underline{\sigma}, \overline{\sigma}], \quad \rho_{ij} \in [-\overline{\rho}, \overline{\rho}].$$

We first work with Σ_δ, fixing x_δ, and note that the program with respect to Σ_δ depends only on $\gamma - \delta$ and $\Pr(\theta = a|f_S)$. Recalling the definition of $\Delta(f_S) = \mathbb{E}[\gamma|f_S, \theta \neq a] - \delta$, the maximizer is equivalent to the maximizer of

$$\max_{\Sigma_\delta} \mathbb{E}^{\delta}_{f_S}\big[\operatorname{tr}(\Delta(f_S)\Sigma_\delta)\big] = \max_{\Sigma_\delta} \mathbb{E}^{\delta}_{f_S}\Big[\sum_{i=1}^{n}\Delta_{ii}(f_S)\sigma_i^2 + \sum_{i<j}(\Delta_{ij} + \Delta_{ji})\rho_{ij}\sigma_i\sigma_j\Big]$$

subject to the same constraints. This program is linear in the objective, meaning the solution is given by an extreme point of the feasible set. In particular, fixing σ_i and σ_j, the objective is linear in ρ_{ij}. Thus, we have

$$\begin{aligned} \mathbb{E}^{\delta}_{f_S}[\Delta_{ij} + \Delta_{ji}] > 0 &\implies \rho_{ij} = \overline{\rho} \\ \mathbb{E}^{\delta}_{f_S}[\Delta_{ij} + \Delta_{ji}] < 0 &\implies \rho_{ij} = -\overline{\rho}, \end{aligned}$$

noting that because $|\rho| < 1$, we have:

$$\det\begin{pmatrix} 1 & \rho_{12} & \cdots & \rho_{1j} \\ \rho_{12} & 1 & \cdots & \rho_{2j} \\ \vdots & & \ddots & \vdots \\ \rho_{1j} & \rho_{2j} & \cdots & 1 \end{pmatrix} > 0$$

We further note that the value of ρ does not depend on σ_i or σ_j, since they are always positive. With the values of ρ_{ij} in hand, we then note that because $\sigma_i > 0$ for all i, we necessarily have that $\prod_{i=1}^{j} \sigma_i^2 > 0$, meaning the positive definiteness constraint is always satisfied strictly. Finally, the optimal choice of σ_i is given by taking the program above, plugging in the values of ρ_{ij}, and solving the linear program, which yields either $\sigma_i = \overline{\sigma}_i$ or $\underline{\sigma}_i$ as maximizers.

the program is linear in the entries of Σ_δ. In particular, the sign of the coefficient on Σ_{ij} is given by the sign of $\Delta_{ij}(f_S) + \Delta_{ji}(f_S)$. □

Proposition 3. *Fix δ. The value of information $V(\delta)$ for the receiver from hiring a type δ sender is given by*

$$V(\delta) = \mathbb{E}^R_{f_R}\left[\mathbb{E}^R_\gamma[\gamma|f_R'(x_\delta)]\left(x^* - \frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right],$$

where

$$x^* = \max\left\{0, \min\left\{\overline{x}, x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f_R'(x_\delta)]}\right\}\right\}$$

Proof. Suppose x^* maximizes $\mathbb{E}^R_{f_R|f_S}[f_R(x)|f_S(x)]$ and x_0 maximizes $\mathbb{E}^R_{f_R}[f_R(x)]$. Conditional on f_S, we have:

$$\begin{aligned}
&\mathbb{E}^R_{f_R|f_S}[f_R(x^*)] - \mathbb{E}^R_{f_R|f_S}[f_R(x_0)] \\
&= f_S(x^*) + \Delta(f_S)\left(\sigma_\delta^2 - (x^* - x_\delta)^2\right) - f_S(x_0) - \Delta(f_S)\left(\sigma_\delta^2 - (x_0 - x_\delta)^2\right) \\
&= \beta_1(x^* - x_0) - \delta\left(x^{*2} - x_0^2\right) + \Delta(f_S)\left((x_0 - x_\delta)^2 - (x^* - x_\delta)^2\right) \\
&= \beta_1(x^* - x_0) - \delta\left(x^{*2} - x_0^2\right) + \Delta(f_S)\left(2x_\delta(x^* - x_0) - (x^{*2} - x_0^2)\right) \\
&= \left(\beta_1 + 2\Delta(f_S)x_\delta\right)(x^* - x_0) - \left(\delta + \Delta(f_S)\right)\left(x^{*2} - x_0^2\right) \\
&= (x^* - x_0)\left[\beta_1 + 2\Delta(f_S)x_\delta - \left(\delta + \Delta(f_S)\right)(x^* + x_0)\right] \\
&= (x^* - x_0)\left[2\left(\delta + \Delta(f_S)\right)x^* - \left(\delta + \Delta(f_S)\right)(x^* + x_0)\right] \\
&= \left(\delta + \Delta(f_S)\right)(x^* - x_0)^2 \\
&= \mathbb{E}^R_\gamma[\gamma|f_S]\,(x^*(f_S) - x_0)^2.
\end{aligned}$$

In turn, we have

$$V(\delta) = \mathbb{E}^R_{f_S|\delta}\left[\mathbb{E}^R_\gamma[\gamma|f_S]\,(x^*(f_S) - x_0)^2\right].$$

In particular, we have $x_0 = \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}$ and, from the previous result,

$$x^* = \frac{\beta_1 + 2\Delta(f_S)x_\delta}{2\mathbb{E}_\gamma^R[\gamma|f_S]} = x_\delta + \frac{\beta_1 - 2\delta x_\delta}{2\mathbb{E}_\gamma^R[\gamma|f_S]}.$$

This leaves us with:

$$V(\delta) = \mathbb{E}_{f_S|\delta}^R\left[\mathbb{E}_\gamma^R[\gamma|f_S]\left(x_\delta + \frac{\beta_1 - 2\delta x_\delta}{2\mathbb{E}_\gamma^R[\gamma|f_S]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2\right].$$

Note moreover that $\beta_1 - 2\delta x_\delta = f_S'(x_\delta) = f_R'(x_\delta)$. This means that we can write the value as:

$$V(\delta) = \mathbb{E}_{f_R}^R\left[\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2\right],$$

□

Proposition 4. *Writing the value of information $V(\delta)$ as a function of the parameter x_δ, $V(\delta, x_\delta)$, we have*

$$V_{x_\delta}(\delta, x_\delta) = -\mathbb{E}_{f_R}^R\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\left(\frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]}\right)^2\right].$$

Moreover if p_1^R is absolutely continuous as defined on all of $\mathbb{R}$ and $\lim_{|\alpha_1|\to\infty} \alpha_1^2 p_1^R(\alpha_1) = 0$, $\arg\max_{x_\delta} V(\delta, x_\delta) > 0$.

Proof. Recalling that $f_R''(x_\delta) = -2\gamma$ and using the previous proposition, we note that

$$\begin{aligned}\frac{\partial V(\delta)}{\partial x_\delta} = \mathbb{E}_{f_R}^R\Bigg[&\left(\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right)^2 \\ &+ 2\,\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\left(x_\delta + \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{\mathbb{E}_1^R[\alpha_1]}{2\mathbb{E}_\gamma^R[\gamma]}\right) \\ &\cdot\left(1 - \frac{\gamma}{\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]} - \frac{f_R'(x_\delta)}{2\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]^2}\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right)\Bigg]\end{aligned}$$

In particular, we note that by Bayes' Rule:

$$\mathbb{E}_{f_R}^R\left[\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right] = \mathbb{E}_\gamma^R[\gamma] \implies \mathbb{E}_{f_R}^R\left[\frac{\partial}{\partial x_\delta}\mathbb{E}_\gamma^R[\gamma|f_R'(x_\delta)]\right] = 0.$$

Using this we simplify the above expression to

$$\begin{aligned}
&\mathbb{E}^R_{f_R}\Bigg[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]\right)\left(x_\delta+\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}-\frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)\\
&\quad\cdot\left(x_\delta+\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}-\frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}-\frac{f'_R(x_\delta)}{\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}\right)\\
&\quad+\underbrace{2\left(x_\delta+\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}-\frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)\cdot\left(\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]-\gamma\right)}_{=0 \text{ conditioning on } f'_R}\Bigg]\\
=&\mathbb{E}^R_{f_R}\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]\right)\left(x_\delta+\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}-\frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)\left(x_\delta-\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}-\frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)\right]\\
=&\mathbb{E}^R_{f_R}\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]\right)\left(\left(x_\delta-\frac{\mathbb{E}^R_1[\alpha_1]}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2-\left(\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}\right)^2\right]\\
=&-\mathbb{E}^R_{f_R}\left[\left(\frac{\partial}{\partial x_\delta}\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]\right)\left(\frac{f'_R(x_\delta)}{2\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]}\right)^2\right].
\end{aligned}$$

Finally, we evaluate this expression at $x_\delta = 0$. In this case, note that $f'_R(0) = \alpha_1$, so the expression simplifies to

$$-\mathbb{E}^R_{f_R}\left[\left(\frac{\partial}{\partial x_\delta}\Big|_{x_\delta=0}\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]\right)\left(\frac{\alpha_1}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right].$$

Next, we express R's posterior mean of γ given $f'_R(x_\delta)$ as

$$\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)] = \frac{\mathbb{E}^R_{\tilde\gamma}\left[\frac{p_1^R(\alpha_1-2\gamma x_\delta+2\tilde\gamma x_\delta)}{p_1^R(\alpha_1)}\tilde\gamma\right]}{\mathbb{E}^R_{\tilde\gamma}\left[\frac{p_1^R(\alpha_1-2\gamma x_\delta+2\tilde\gamma x_\delta)}{p_1^R(\alpha_1)}\right]}$$

Differentiating with respect to x_δ yields

$$\frac{\mathbb{E}^R_{\tilde\gamma}\left[\frac{2(\tilde\gamma-\gamma)p_1^{R'}(\alpha_1-2\gamma x_\delta+2\tilde\gamma x_\delta)}{p_1^R(\alpha_1)}\tilde\gamma\right]}{\mathbb{E}^R_{\tilde\gamma}\left[\frac{p_1^R(\alpha_1-2\gamma x_\delta+2\tilde\gamma x_\delta)}{p_1^R(\alpha_1)}\right]}-\frac{\mathbb{E}^R_{\tilde\gamma}\left[\frac{2(\tilde\gamma-\gamma)p_1^{R'}(\alpha_1-2\gamma x_\delta+2\tilde\gamma x_\delta)}{p_1^R(\alpha_1)}\right]\mathbb{E}^R_{\tilde\gamma}\left[\frac{p_1^R(\alpha_1-2\gamma x_\delta+2\tilde\gamma x_\delta)}{p_1^R(\alpha_1)}\tilde\gamma\right]}{\mathbb{E}^R_{\tilde\gamma}\left[\frac{p_1^R(\alpha_1-2\gamma x_\delta+2\tilde\gamma x_\delta)}{p_1^R(\alpha_1)}\right]^2}.$$

But at $x_\delta = 0$ this can be heavily simplified to

$$\begin{aligned}
&\mathbb{E}^R_{\tilde\gamma}\Big[\frac{2(\tilde\gamma-\gamma)p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)}\tilde\gamma\Big] - \mathbb{E}^R_{\tilde\gamma}\Big[\frac{2(\tilde\gamma-\gamma)p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)}\Big]\mathbb{E}^R_{\tilde\gamma}\big[\tilde\gamma\big] \\
&= \frac{2p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)}\Big(\mathbb{E}^R_{\tilde\gamma}\big[\tilde\gamma(\tilde\gamma-\gamma)\big] - \mathbb{E}^R_{\tilde\gamma}\big[(\tilde\gamma-\gamma)\big]\mathbb{E}^R_{\tilde\gamma}\big[\tilde\gamma\big]\Big) \\
&= \frac{2p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)}\mathrm{Var}^R_{\tilde\gamma}(\tilde\gamma)
\end{aligned}$$

Plugging this into the original equation yields

$$\begin{aligned}
\left.\frac{\partial V(\delta)}{\partial x_\delta}\right|_{x_\delta=0} &= -\mathbb{E}^R_{f_R}\left[\frac{2p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)}\mathrm{Var}^R_{\tilde\gamma}(\tilde\gamma)\left(\frac{\alpha_1}{2\mathbb{E}^R_\gamma[\gamma]}\right)^2\right] \\
&= -2\mathbb{E}^R_{f_R}\left[\frac{p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)}\alpha_1^2\right]\mathbb{E}^R_{f_R}\left[\frac{\mathrm{Var}^R_{\tilde\gamma}(\tilde\gamma)}{\mathbb{E}^R_\gamma[\gamma]^2}\right]
\end{aligned}$$

But note that using integration by parts, the first term can be expressed as:

$$\mathbb{E}^R_{f_R}\left[\frac{p_1^{R'}(\alpha_1)}{p_1^R(\alpha_1)}\alpha_1^2\right] = \int_{\mathbb{R}}\alpha_1^2 p_1^{R'}(\alpha_1)d\alpha_1 = \alpha_1^2 p_1^R(\alpha_1)\Big|_{-\infty}^{+\infty} - \int_{\mathbb{R}}2\alpha_1 p_1^R(\alpha_1)d\alpha_1 = -2\mathbb{E}^R_1[\alpha_1].$$

This shows that the derivative of $V(\delta)$ is *positive* at $x_\delta = 0$, and hence that its maximizer is strictly positive. □

Proposition 5. *Fix a sender type δ and let $x^*(f_S)$ maximize $\mathbb{E}_{f_R|f_S,\mu_\delta}[y|x]$. Then, there exists $\sigma^*_\delta \in (\underline{\sigma},\overline{\sigma})$ that maximizes*

$$W(\sigma_\delta) := \mathbb{E}^R_{f_S|\delta}\Big[\mathrm{Var}^R_{f_R|f_S}\big(y|x^*(f_S),x_\delta,\sigma_\delta\big)\Big].$$

with respect to σ_δ. Moreover, W is strictly decreasing in σ for $\sigma < \sigma^*_\delta$ and increasing for $\sigma > \sigma^*_\delta$.

Proof. With some abuse of notation, given f_S and the other parameters, we have:

$$\begin{aligned}
\mathrm{Var}(y|x^*) &= \mathrm{Var}\big(f_S(x^*(f_S)) + (\gamma-\delta)(\sigma_\delta^2 - (x^*(f_S)-x_\delta)^2)\big) \\
&= \big((\sigma_\delta^2 - (x^*(f_S)-x_\delta)^2)\big)^2\mathrm{Var}(\gamma|f_S,x_\delta),
\end{aligned}$$

where we note that information about γ depends on both f_S as well as x_δ. Because the sender's model fixes curvature at a single value, R remains uncertain about γ after seeing f_S, and this variance is > 0. Hence, our problem is to choose x_δ and σ_δ to minimize

$$\mathbb{E}^R_{f_S|\delta}\Big[\big((\sigma_\delta^2 - (x^*(f_S)-x_\delta)^2)\big)^2\mathrm{Var}(\gamma|f_S,x_\delta)\Big].$$

We note that this function is quadratic in σ_δ^2 and minimized at

$$\sigma_\delta^2 = \frac{\mathbb{E}^R_{f_S|\delta}\big[(x^*(f_S) - x_\delta)^2)\mathrm{Var}(\gamma|f_S, x_\delta)\big]}{\mathbb{E}^R_{f_S|\delta}\big[\mathrm{Var}(\gamma|f_S, x_\delta)\big]} = \frac{\mathbb{E}^R_{f_S|\delta}\big[\big(\frac{\beta_1 - 2\delta x_\delta}{2\mathbb{E}^R_\gamma[\gamma|f_S]}\big)^2\mathrm{Var}(\gamma|f_S, x_\delta)\big]}{\mathbb{E}^R_{f_S|\delta}\big[\mathrm{Var}(\gamma|f_S, x_\delta)\big]}$$

The terms in the numerator represent the ex-ante expectation of the squared difference between x_δ and the eventual choice $x^*(f_S)$ multiplied by the variance of γ. The term in the denominator is the expected variance of γ. Because there is uncertainty about $x^*(f_S)$, the numerator is always positive and bounded away from 0, which gives the result.

□

Proposition 6. *Suppose R chooses $x^*(f_S)$ to maximize (4). Then:*

1. *A naive sender chooses $\sigma_\delta > \sigma_\delta^*$ and a cautious sender chooses $\sigma_\delta < \sigma_\delta^*$*
2. *As $\psi \to 1$, $x^*(f_S) \to x_\delta \pm \sigma$.*

Proof. Given f_S and δ and using the previous result, R's problem is to maximize

$$\max_x (1-\psi)\big(f_S(x) + \Delta(f_S)(\sigma_\delta^2 - (x - x_\delta)^2)\big) - \psi\big((\sigma_\delta^2 - (x - x_\delta)^2)\big)^2\mathrm{Var}(\gamma|f_S),$$

which S then integrates out over the plausible values of f_S. Denote this maximizer by $x^*(f_S)$. By the envelope theorem, the derivative with respect to σ_δ^2 is

$$\mathbb{E}_{f_S}\big[(1-\psi)\Delta(f_S) - 2\psi\big(\sigma_\delta^2 - (x^*(f_S) - x_\delta)^2\big)\mathrm{Var}(\gamma|f_S)\big].$$

Recall from above that the expectation of $\big(\sigma_\delta^2 - (x^*(f_S) - x_\delta)^2\big)\mathrm{Var}(\gamma|f_S)$ is positive below σ_δ^* and negative above. For a naive sender, $\Delta(f_S) > 0$, meaning the derivative is still positive at σ_δ^*, and thus that his choice of σ is higher. The opposite holds for the naive sender.

The second result follows by noting that as $\psi \to 1$, R is simply choosing x^* to minimize variance. But this can conditionally always be achieved by setting $x^* = x_\delta \pm \sigma_\delta$, which gives the result. □

Theorem 3. *Suppose P_δ^a is continuous. Then, the optimal type space is $\Theta = [0, \overline{\gamma}] = \Gamma$. Moreover, there exists a function $\pi(\delta)$ such that T is given by a continuous distribution t with*

$$t(\delta) = p_\delta^a(\delta)\pi(\delta). \tag{T^*}$$

In particular, there exist thresholds $0 \le \underline{\delta}_0 < \mathbb{E}^R_\gamma[\gamma] < \overline{\delta}_0 \le \overline{\gamma}$ such that $\pi(\delta)$ is strictly decreasing below $\underline{\delta}_0$, strictly increasing above $\overline{\delta}_0$, and equal to 0 on $[\underline{\delta}_0, \overline{\delta}_0]$.

Proof. Consider a sender model $f_S = \beta_0 + \beta_1 x - \delta x^2$. First, we show that $\Theta = \Gamma$. For each $\delta \in \Gamma$, a can create a type δ. This type strategically samples by setting σ such that $\max_x f_S(x) \le \max_x f_S(x) + (\mathbb{E}[\gamma|\delta])(\sigma^2 - (x - x_\delta)^2)$. Thus, conditional on seeing a model of the form $\beta_0 + \beta_1 x - \delta x^2$, R believes

that $f_S(x)$ is either the true model or the model from the type δ sender. a's utility conditional on providing this model increases from the existence of the type δ sender whenever $\delta \neq \mathbb{E}^R_\gamma[\gamma]$, and the gain is increasing in the posterior probability that R places on the type δ sender. Hence, $\Theta = \Gamma$.

We now determine the distribution of types T. Fix a model $\beta_0+\beta_1 x-\delta x^2$. First, for a distribution $T \in \mathcal{P}([0,\overline{\gamma}])$ (the set of measures over $[0,\overline{\gamma}]$), we write as $q(T,\delta)$ the posterior probability that this model came from a type δ sender.

$$q(T,\delta) = \begin{cases} \frac{(1-p_a)t(\delta)}{(1-p_a)t(\delta)+p_a p^a_\delta(\delta)} & T \text{ admits density } t(\delta) \\ 1 & T(\{\delta\}) \text{ has a mass point} \end{cases}$$

With some abuse of notation we write as $\mathbb{E}^\gamma_R[\gamma]$ R's expectation given the sender was a simple type who sampled at x_δ. meaning that

$$\mathbb{E}^R_\gamma[\gamma|f_S] = q(T,\delta)\mathbb{E}^R_\gamma[\gamma] + \big(1-q(T,\delta)\big)\delta.$$

Given a model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$, the receiver's maximal expected value is

$$f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma^2_\delta - (x^*-x_\delta)^2\big)$$

where x^* is a function of β_1, δ, and $\mathbb{E}^R_\gamma[\gamma|f_S]$, and σ_δ and x_δ are determined as in Theorem 1.

Our objective can thus be rewritten as:

$$\max_{T\in\mathcal{P}([0,\overline{\gamma}])} \int_{\mathbb{R}}\int_{\mathbb{R}}\int_0^{\overline{\gamma}} f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma^2_\delta - (x^*-x_\delta)^2\big) dP^a_\delta(\delta) dP^a_1(\beta_1) dP^a_0(\beta_0),$$

Note that because P^a_δ is continuous, there are no gains to placing a positive mass on any singleton δ value. Hence, our problem boils down to

$$\max_{t(\delta)} \int_{\mathbb{R}}\int_{\mathbb{R}}\int_0^{\overline{\gamma}} f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma^2_\delta - (x^*-x_\delta)^2\big) dP^a_\delta(\delta) dP^a_1(\beta_1) dP^a_0(\beta_0)$$
$$\text{s.t. } t(\delta) \geq 0 \text{ and } \int_0^{\overline{\gamma}} t(\delta) d\delta = 1.$$

Using a calculus of variations approach, we can write a Lagrangian as:

$$\mathcal{L}(t,\lambda) = \int_{\mathbb{R}}\int_{\mathbb{R}}\int_0^{\overline{\gamma}} f_S(x^*) + q(T,\delta)\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma^2_\delta - (x^*-x_\delta)^2\big) p^a_\delta(\delta) + \lambda\Big(\frac{1}{\overline{\gamma}} - t(\delta)\Big) d\delta dP^a_1(\beta_1) dP^a_0(\beta_0).$$

Using the envelope theorem, we obtain the following first-order condition.

$$\mathbb{E}^a_{0,1}\left[\frac{(1-p_a)\big((1-p_a)t(\delta)+p_a p^a_\delta(\delta)\big) - (1-p_a)^2 t(\delta)}{((1-p_a)t(\delta)+p_a p^a_\delta(\delta))^2}\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma^2_\delta-(x^*-x_\delta)^2\big)p^a_\delta(\delta)\right] - \lambda$$
$$= \frac{(1-p_a)p_a p^a_\delta(\delta)}{((1-p_a)t(\delta)+p_a p^a_\delta(\delta))^2} p^a_\delta(\delta)\big(\mathbb{E}^R_\gamma[\gamma]-\delta\big)\big(\sigma^2_\delta - \mathbb{E}^a_1\big[(x^*-x_\delta)^2\big]\big) - \lambda,$$

where we note that x^* depends on β_1 and δ. Now, define

$$P = p_a p_\delta^a(\delta) \qquad R = (1-p_a)t(\delta).$$

Recall that $x^* = x_\delta + \frac{\beta_1 - 2\delta x_\delta}{2\mathbb{E}_\gamma^R[\gamma|\delta]}$. We then have

$$\begin{aligned}
\frac{(1-p_a)P^2}{p_a(P+R)^2}\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\sigma_\delta^2 - \mathbb{E}_1^a\big[(x^*-x_\delta)^2\big]\big) &= \frac{(1-p_a)P^2}{p_a(P+R)^2}\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\sigma_\delta^2 - \mathbb{E}_1^a\big[\big(\frac{\beta_1-2\delta x_\delta}{2\mathbb{E}_\gamma^R[\gamma|\delta]}\big)^2\big]\big) \\
&= \frac{(1-p_a)P^2}{p_a(P+R)^2}\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\sigma_\delta^2 - \frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{\frac{4(P\delta+R\mathbb{E}_\gamma^R[\gamma])^2}{(P+R)^2}}\big) \\
&= \frac{(1-p_a)}{p_a}P^2\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(P+R)^2} - \frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{4(P\delta+R\mathbb{E}_\gamma^R[\gamma])^2}\big).
\end{aligned}$$

Next, define $\pi(\delta) := \frac{t(\delta)}{p_\delta^a(\delta)}$ and $c = \frac{1-p_a}{p_a}$. In particular, $\frac{R}{P} = c\pi(\delta)$. This definition gives:

$$\begin{aligned}
&\frac{(1-p_a)}{p_a}P^2\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(P+R)^2} - \frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{4(P\delta+R\mathbb{E}_\gamma^R[\gamma])^2}\big) \\
&= c\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^2}\big).
\end{aligned}$$

Finally, we verify the second order condition; because $\pi(\delta) = \frac{t(\delta)}{p_\delta^a(\delta)}$, differentiating with respect to t yields:

$$c^2 p_\delta^a(\delta)\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(-\frac{\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}_\gamma^R[\gamma]\frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{2(\delta+c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^3}\big).$$

For $\delta < \mathbb{E}_\gamma^R[\gamma]$, we note that $\sigma_\delta \to \overline{\sigma}$, meaning $-\frac{\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}_\gamma^R[\gamma]\frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{2(\delta+c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^3} < 0$. For $\delta > \mathbb{E}_\gamma^R[\gamma]$, $\sigma_\delta \to \underline{\sigma}$ small, meaning $-\frac{\sigma_\delta^2}{(1+c\pi(\delta))^3} + \mathbb{E}_\gamma^R[\gamma]\frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{2(\delta+c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^3} > 0$, and thus that the whole expression is negative. Hence, the original objective is concave in t, meaning the first order condition identifying a solution is given by

$$c\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^2}\big) = \lambda.$$

Note that the left hand side is strictly positive except at $\delta = \mathbb{E}_\gamma^R[\gamma]$, when it is equal to 0. We define:

$$\xi(\delta,\pi) := c\big(\mathbb{E}_\gamma^R[\gamma]-\delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi)^2} - \frac{\mathbb{E}_1^a[(\beta_1-2\delta x_\delta)^2]}{4(\delta+c\pi\mathbb{E}_\gamma^R[\gamma])^2}\big).$$

By the Implicit Function Theorem, we have $\pi'(\delta) = -\frac{\xi_\pi}{\xi_\delta}$. But note that $\xi_\pi < 0$ (since the objective was concave in t, and $t = p_\delta^a(\delta)\cdot\pi$), meaning that the sign of π' is just the sign of ξ_δ.

Now, consider $\delta < \mathbb{E}_\gamma^R[\delta]$. These senders sample so that $\sigma_\delta = \overline{\sigma}$ and $x_\delta = \max\{0, \min\{\frac{\mathbb{E}_1^a[\beta_1]}{2\delta}, \overline{x}\}\}$. Consider first the interior case, which gives

$$\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)^2] = \mathbb{E}_1^a[(\beta_1 - 2\delta\frac{\mathbb{E}_1^a[\beta_1]}{2\delta})^2] = \mathrm{Var}_1^a[\beta_1]$$
$$\implies \xi(\delta, \pi) = c\big(\mathbb{E}_\gamma^R[\gamma] - \delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi)^2} - \frac{\mathrm{Var}_1^a[\beta_1]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\big).$$

The derivative of this expression with respect to δ is < 0 if and only if

$$-c\big(\frac{\sigma_\delta^2}{(1+c\pi)^2} - \frac{\mathrm{Var}_1^a[\beta_1]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\big) + c\big(\mathbb{E}_\gamma^R[\gamma] - \delta\big)\frac{\mathrm{Var}_1^a[\beta_1]}{2(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^3} < 0$$
$$\frac{\sigma_\delta^2}{(1+c\pi)^2} \geq \frac{\mathrm{Var}_1^a[\beta_1]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\left(1 + \frac{2(\mathbb{E}_\gamma^R[\gamma] - \delta)}{\delta + c\pi\mathbb{E}_\gamma^R[\gamma]}\right)$$

which always holds for $\sigma_\delta \to \overline{\sigma}$ sufficiently large. We are left to consider the case of a corner solution (0 or $\overline{x}$). The derivative ξ with respect to δ in this case is < 0 if and only if

$$-\big(\frac{\sigma_\delta^2}{(1+c\pi)^2} - \frac{\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\big) + \big(\mathbb{E}_\gamma^R[\gamma] - \delta\big)\left(\frac{\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)^2]}{2(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^3} + \frac{x_\delta\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)]}{(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\right) < 0,$$

which again holds for σ_δ^2 large. Thus, $\pi(\delta)$ is decreasing in δ for $\delta < \mathbb{E}_\gamma^R[\gamma]$.

Now, we consider $\delta > \mathbb{E}_\gamma^R[\gamma]$. In this case, we have $\sigma_\delta \to 0$. Moreover, we have that $x_\delta = \arg\max_{x\in[0,\overline{x}]}\mathbb{E}_1^a[(\beta_1 - 2\delta x)^2]$ (i.e. specialist senders sample wherever they expect the curve to be steepest). In this case, we have

$$\xi(\delta, \pi) \approx c\big(\delta - \mathbb{E}_\gamma^R[\gamma]\big)\frac{\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi\mathbb{E}_\gamma^R[\gamma])^2}\big).$$

$\delta - \mathbb{E}_\gamma^R[\gamma]$ is increasing in δ (and positive for $\delta > \mathbb{E}_\gamma^R[\gamma]$). Using the fact that x_δ maximizes $\mathbb{E}_1^a[(\beta_1 - 2\delta x)^2]$, the second term is also increasing in δ for x_δ extreme.

Thus, whenever $\pi(\delta) > 0$, it is strictly decreasing for $\delta < \mathbb{E}_\gamma^R$ and strictly increasing for $\delta > \mathbb{E}_\gamma^R$. To see that $\pi(\delta) = 0$ on an interior interval we simply note that for every $\lambda > 0$, there exists an interval of δ values containing $\mathbb{E}_\gamma^R[\gamma]$ such that

$$c\big(\mathbb{E}_\gamma^R[\gamma] - \delta\big)\big(\frac{\sigma_\delta^2}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}_1^a[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi(\delta)\mathbb{E}_\gamma^R[\gamma])^2}\big) < \lambda$$

at $\pi = 0$ (and, by concavity, for all $\pi \geq 0$).

□

Proposition 7. *The polarization of the type space is increasing in p_a. That is, $\underline{\delta}_0$ is decreasing in p_a and $\overline{\delta}_0$ is increasing in p_a. As $p_a \to 1$, $t(\delta)$ converges to a distribution that either places all mass only on $\delta = 0$, $\delta = \overline{\gamma}$, or both. As $p_a \to 0$, $t(\delta)$ converges to a distribution that is strictly decreasing on $[0, \mathbb{E}_\gamma^R[\gamma])$, equal*

to 0 *at* $\mathbb{E}^R_\gamma[\gamma]$*, and strictly increasing on* $(\mathbb{E}^R_\gamma[\gamma], \overline{\gamma}]$.

Proof. We recall our initial first order condition implicitly defining $\pi(\delta)$

$$c\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\Big(\frac{\sigma^2_\delta}{(1+c\pi(\delta))^2} - \frac{\mathbb{E}^a_1[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi(\delta)\mathbb{E}^R_\gamma[\gamma])^2}\Big) = \lambda,$$

where the left hand side is decreasing in π, decreasing in δ for $\delta < \mathbb{E}^R_\gamma[\gamma]$, and increasing in δ for $\delta > \mathbb{E}^R_\gamma[\gamma]$. We also recall that $c := \frac{1-p_a}{p_a}$, and that as $c \to 0$, $p_a \to 1$, and that as $c \to \infty$, $p_a \to 0$.

Recall also that $\overline{\delta}_0$ and $\underline{\delta}_0$ are the points at which $\pi(\delta) = 0$ and are, in turn, functions of c. By the constraint that $t(\delta)$ is a density and by writing $\pi(\delta, c)$ as a function of c, we have

$$\int_0^{\overline{\gamma}} t(\delta)d\delta = \int_0^{\underline{\delta}_0(c)} \pi(\delta,c)p^a_\delta(\delta)d\delta + \int_{\overline{\delta}_0(c)}^{\overline{\gamma}} \pi(\delta,c)p^a_\delta(\delta)d\delta = 1$$

$$\pi(\overline{\delta}_0(c),c)p^a_\delta(\overline{\delta}_0(c))\overline{\delta}_0'(c) - \pi(\underline{\delta}_0(c),c)p^a_\delta(\underline{\delta}_0(c))\underline{\delta}_0'(c) + \int_0^{\underline{\delta}_0(c)} \pi_c(\delta,c)p^a_\delta(\delta) + \int_{\overline{\delta}_0(c)}^{\overline{\gamma}} \pi_c(\delta,c)p^a_\delta(\delta)$$
$$= \int_0^{\overline{\gamma}} \pi_c(\delta,c)p^a_\delta(\delta)d\delta = 0,$$

where we use the fact that $\pi(\overline{\delta}_0(c), c) = \pi(\underline{\delta}_0(c), c) = 0$.

Next, note that, fixing δ, the first order condition can be rearranged as

$$\Xi(c\pi, \delta) := \big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\Big(\frac{\sigma^2_\delta}{(1+c\pi)^2} - \frac{\mathbb{E}^a_1[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta + c\pi\mathbb{E}^R_\gamma[\gamma])^2}\Big) = \frac{\lambda(c)}{c}$$

The FOC then gives

$$\Xi_{(c\pi)}(c\pi_c + \pi) = \frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2} \implies c\pi_c = \frac{\frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2}}{\Xi_{(c\pi)}} - \pi$$

But then, using the boundary condition, we have

$$c\int_0^{\overline{\gamma}} \pi_c(\delta,c)p^a_\delta(\delta)d\delta = \int_0^{\overline{\gamma}} \frac{\frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2}}{\Xi_{(c\pi)}} p^a_\delta(\delta) - \pi(\delta,c)p^a_\delta(\delta)d\delta = 0$$
$$\implies \left(\lambda'(c) - \frac{\lambda(c)}{c}\right)\int_0^{\overline{\gamma}} \frac{1}{\Xi_{(c\pi)}(c\pi,\delta)} p^a_\delta(\delta)d\delta = c$$
$$\implies \frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2} = \frac{1}{\int_0^{\overline{\gamma}} \frac{1}{\Xi_{(c\pi)}(c\pi,\delta)} p^a_\delta(\delta)d\delta} < 0$$

since $\Xi_{(c\pi)} < 0$ (i.e. Ξ is decreasing in $c\pi$, from the proof of the previous Theorem).

Now, consider $\Xi(0, \underline{\delta}_0(c))$, which implicitly defines $\underline{\delta}_0(c)$ as

$$\Xi(0, \underline{\delta}_0(c)) := \big(\mathbb{E}^R_\gamma[\gamma] - \underline{\delta}_0(c)\big)\Big(\sigma^2_\delta - \frac{\mathbb{E}^a_1[(\beta_1 - 2\underline{\delta}_0(c)x_\delta)^2]}{4(\underline{\delta}_0(c))^2}\Big) = \frac{\lambda(c)}{c}.$$

In this case, we have

$$\Xi_\delta(0, \underline{\delta}_0(c))\underline{\delta}_0'(c) = \frac{\lambda'(c)}{c} - \frac{\lambda(c)}{c^2} < 0.$$

But recall from the proof of the previous result that $\Xi_\delta < 0$ for $\delta < \mathbb{E}^R_\gamma[\gamma]$. This means that $\underline{\delta}_0(c)$ is increasing in c, i.e. is increasing in $\frac{1-p_a}{p_a}$. Hence, as p_a increases, $\underline{\delta}_0$ decreases. An analogous proof shows that as p_a increases, $\overline{\delta}_0$ increases. That is, the "polarization" induced by π is increasing in p_a.

Finally, we investigate the limiting cases as $p_a \to 0$ ($c \to \infty$) and $p_a \to 1$ ($c \to 0$). We begin with $c \to \infty$, and claim that $\underline{\delta}_0 \to \mathbb{E}^R_\gamma[\gamma]$. First, by our work above $\frac{\lambda(c)}{c}$ is decreasing in c, meaning it converges to a limit. We claim $\lim_{c\to\infty} \frac{\lambda(c)}{c} = 0$. Suppose by contradiction that this did not hold, and instead it converged to some $k > 0$. This would mean that there exists $\overline{c\pi} > 0$ such that $c\pi(\delta) \leq \overline{c\pi}$ for all c and $\delta \in [0, \overline{\gamma}] - \lim_{c\to\infty}[\underline{\delta}_0(c), \overline{\delta}_0(c)]$. But then, for all c we would have

$$\int_0^{\overline{\gamma}} \pi(\delta, c) p^a_\delta(\delta) d\delta \leq \int_0^{\overline{\gamma}} \overline{c\pi} p^a_\delta(\delta) d\delta = \frac{\overline{c\pi}}{c},$$

which approaches 0 as $c \to \infty$, contradicting that $\int_0^{\overline{\gamma}} \pi(\delta, c) p^a_\delta(\delta) d\delta = 1$. Hence, we have that $\lim_{c\to\infty} \frac{\lambda(c)}{c} = 0$, meaning $\lim_{c\to\infty} \underline{\delta}_0(c) = \lim_{c\to\infty} \overline{\delta}_0(c) = \mathbb{E}^R_\gamma[\gamma]$. Moreover, we note that even for c arbitrarily large, $\pi(\mathbb{E}^R_\gamma[\gamma]) = 0$, with $\pi'(\delta) < 0$ below $\mathbb{E}^R_\gamma[\gamma]$ and $\pi'(\delta) > 0$ above $\mathbb{E}^R_\gamma[\gamma]$ since $\Xi(c\pi, \delta)$ is decreasing in δ and increasing in δ on each range, respectively.

Finally, we investigate the case where $c \to 0$. We claim that as $c \to 0$, $\underline{\delta}_0 \to 0$ and $\overline{\delta}_0 \to \overline{\gamma}$ (and hence t converges to a discrete distribution that either concentrates mass on 0 or $\overline{\gamma}$ or both). In particular, note that for all c, we have

$$c = \int_0^{\overline{\gamma}} c\pi(\delta, c) p^a_\delta(\delta) d\delta,$$

which means that as $c \to 0$, we necessarily must have $c\pi(\delta, c) \to 0$ as well. This can happen only if

$$\big(\mathbb{E}^R_\gamma[\gamma] - \delta\big)\big(\sigma^2_\delta - \frac{\mathbb{E}^a_1[(\beta_1 - 2\delta x_\delta)^2]}{4(\delta)^2}\big) \leq \lim_{c\to 0} \frac{\lambda(c)}{c},$$

i.e. either the multiplier constraint must hold with equality or fail to hold (generating a corner solution at $\pi(\delta, c) = 0$). Since this must hold pointwise for each δ for c sufficiently small, we thus have that $\underline{\delta}_0 \to 0$ and $\overline{\delta}_0 \to \overline{\gamma}$, giving the result. □

B Interval Types and Non-Quadratic Tradeoffs

This appendix relaxes two assumptions from the main text. Sections B.1-B.4 allow sender types to be an *interval* of curvature values, rather than a singleton, thus allowing senders to sometimes correctly estimate tradeoffs, but otherwise neglecting or exaggerating them. Appendix B.5 allows

tradeoffs to be non-quadratic.

B.1 Interval Type Spaces

In the main text, a type δ sender's model fixes curvature at a single value δ. However, in reality, senders and receivers alike may consider multiple tradeoffs in different ways.

- Senders could estimate *some* tradeoffs and assume the worst-case or neglect others. In our example, the consultant may thoroughly estimate how carbon taxes affect energy prices, but assume a worst-case scenario for their effects on political backlash.
- Senders could *calibrate* the value of a tradeoff at a fixed parameter, without estimating it on the real data. For example, the consultant may calibrate the effect of carbon taxes on energy prices using an elasticity from the literature, rather than organically estimating it on data.

Accordingly, I now place the following structure on type spaces. I now assume that a sender's type θ is a *subset* of Γ.

Assumption 2. *For each $\theta \in \Theta$, $\theta \subseteq \Gamma$ is a (potentially degenerate) closed interval. For each $\theta \neq \theta' \in \Theta$, $\theta \cap \theta' = \varnothing$.*

In turn, I now assume a type θ sender provides a models in the set

$$F_\theta : \{\beta_0 + \beta_1 x - \delta x^2 : (\beta_0 1, \beta_1) \in \mathbb{R}^2, \delta \in \theta\}$$

Assumption 2 states that each type corresponds to an interval within $[0, \overline{\gamma}]$. I initially assume type intervals do not intersect for expositional purposes.[19] We will denote $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$. Accordingly, type spaces have the following interpretations.

- If $\overline{\delta}_\theta < \overline{\gamma}$, then a type θ S is *neglecting at least some tradeoffs*. Intuitively, these senders do not allow for the case where γ is close to $\overline{\gamma}$.
- If $\underline{\delta}_\theta > 0$, then a type θ S is *exaggerating at least some tradeoffs*. These senders do not allow for the case where γ is close to 0.

In addition to priors over β_0 and β_1, a type θ sender now has a prior over δ given by $\delta \sim P^\theta_\delta$ with support on $\theta \subseteq \Gamma$. I make the following assumption on P^θ_δ.

Assumption 3. *Suppose $\theta \neq \{0\}$ or $\{\overline{\gamma}\}$. Then, $P^\theta_\delta(\partial\theta \cap (0, \overline{\gamma})) > 0$.*

I impose Assumption 3 as a regularity condition; it is consistent with either P_δ being a discrete distribution, or P_δ being a continuous CDF over $[0, \overline{\gamma}]$ whose mass is censored. At a mathematical level, a type θ S's sampling decision will be determined by how he considers tradeoffs on the "edge" of the interval θ. This "edge" must occur with non-measure-zero probability to have an effect on S's ex-ante expected utility. We denote by $P^\theta_{f_S}$ a type θ sender's prior over models f_S. Second-order beliefs about R's priors over parameters are the same as before.

[19] The case where they do is of particular interest, and I study it in Appendix B.2.

Learning with Interval Types The characterization of $\phi(f_S; \mu_\theta)$ — the set of models R considers in addition to f_S — is the same as above. However, we note that if θ is non-degenerate, we may also have a moment condition corresponding to δ:

$$\mathbb{E}_{\mu_\theta}\left[x^2\left(f_R(x) - \beta_0 - \beta_1 x + \delta x^2\right)\right] \tag{6}$$

This equation may be zero or nonzero, depending on S's type. With interval types, $\Delta(f_S)$ is defined as in the main text. In particular, if $\delta \in \theta^\circ$, we necessarily have that (6) is equal to 0, and that f_S is the true model. In this case, $\Delta = 0$. However, if $\delta \in \partial\theta$, then the receiver may entertain additional models.

The statement of Proposition 1 is identical in the case with interval types, and Corollary **??** continues to hold. This generalizes the intuitions in the two-type model. If $\delta = \overline{\delta}_\theta < \overline{\gamma}$, then $\Delta(f_S) > 0$: R considers models that have steeper curvature than the one S has provided because he has neglected some tradeoffs. If $\delta = \underline{\delta}_\theta > 0$, then $\Delta(f_S) < 0$: R considers models that have shallower curvature than the one S has provided because he has exaggerated some tradeoffs. In the special case where θ is a singleton, R is completely uncertain as to the value of γ, and $\Delta(f_S)$ may be positive or negative.

Strategic Sampling with Interval Types The definitions of naive and cautious models are as in the main text, and the main result on strategic sampling continues to hold.

Proposition 8. *Suppose Assumptions 2 and 3 hold. Then, the conclusions of Theorem 1 hold.*

Theorem 1 can be interpreted as follows. If $\delta \in (\underline{\delta}_\theta, \overline{\delta}_\theta)$, then R believes S's model is correct. However, if $\delta = \overline{\delta}_\theta$, R will consider models with $\gamma \geq \overline{\delta}_\theta$. Thus, as in the two-type case, this creates an incentive for him to sample from all across the choice space. When $\delta = \underline{\delta}_\theta$, the opposite happens: R considers models with $\gamma \leq \underline{\delta}_\theta$, i.e. "flatter" than those that S provided. Similarly to the two-type case, this gives him an incentive to only estimate models in a small neighborhood of x_θ, i.e. to position himself as a "specialist."

Proof. Following from Proposition 1, let $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$. Suppose $\theta^\circ \neq \varnothing$ and $\delta \in \theta^\circ$. Then (6) is necessarily satisfied, and $\phi(f_S; \theta, \mu_\theta) = \{f_S\}$. However, if $\delta = \underline{\delta}_\theta$ or $\overline{\delta}_\theta$, then (6) is a corner solution from the receiver's perspective. If $\delta = \overline{\delta}_\theta$, then any $\gamma \geq \overline{\delta}_\theta$ is plausible from the receiver's perspective. If $\delta = \underline{\delta}_\theta$, any $\gamma \leq \underline{\delta}_\theta$ is plausible. Finally, if $\underline{\delta}_\theta = \overline{\delta}_\theta$, any $\gamma \in [0, \overline{\gamma}]$ value is possible. This pins down the plausible set of γ values from the receiver's perspective.

Next, fix σ. The receiver chooses $x = x^*$ to maximize:

$$\begin{aligned} &\beta_0 + \beta_1 x - \delta x^2 + \left(\mathbb{E}^R_\gamma[\gamma | f_S] - \delta\right)\left(\sigma^2 - (x - x_\theta)^2\right) \\ &\quad = \beta_0 + \beta_1 x - \delta x^2 + \Delta(f_S)\left(\sigma^2 - (x - x_\theta)^2\right) \end{aligned} \tag{7}$$

We note that because the sender behaves as if the receiver has a flat prior over α_0, her posterior places the same mass on all models that are the same "up to a constant"; hence, changing σ^2

does not change the relative mass she places on each model, fixing a plausible γ value. Thus, by the Envelope Theorem, (7) is increasing in σ^2 if $\Delta(f_S)$ is positive and decreasing if $\Delta(f_S)$ is negative. Working backwards, when the sender takes expectations over (7), he chooses $\sigma = \overline{\sigma}$ if $\mathbb{E}_{f_S}[\Delta(f_S)] \geq 0$ and $\underline{\sigma}$ otherwise.

Finally, we characterize the equilibrium value of x_θ. Note that because the sender behaves as if the receiver's prior is flat over α_0 and α_1, his expectation of the receiver's expectation of γ does not depend on x_θ. Returning to (7) and using the envelope theorem, we have that the derivative with respect to x_θ is:

$$2\Delta(f_S)(x^* - x_\theta)$$

This is (strictly) concave if $\Delta(f_S) > 0$, and achieves a maximum when $x^* = x_\theta$. If $\Delta(f_S) \leq 0$, the function is convex and achieves a maximum at a corner solution 0 or $\overline{x}$, giving the potential values for x_θ. Given S must commit to data collection beforehand and integrating backwards, this gives $x_\theta = \mathbb{E}^\theta_{f_S}[x^*]$ if he expects $\Delta(f_S) > 0$ and $x_\theta = 0$ or $\overline{x}$ otherwise.

Finally, we note that $x^*(f_S)$ solves

$$\begin{aligned}\beta_1 - 2\delta x^* - 2\Delta(f_S)(x^* - x_\theta) = 0 &\implies x^* = \frac{\beta_1 + 2\Delta(f_S)x_\theta}{2\mathbb{E}^R_\gamma[\gamma|f_S,\theta]} \\ &\implies \mathbb{E}^\theta_{f_S}[x^*(f_S)] = \mathbb{E}^\theta_{1,\delta}\left[\frac{\beta_1 + 2\Delta(f_S)x_\theta}{2\delta + 2\Delta(f_S)}\right].\end{aligned}$$

□

The multidimensional results of Section 3 extend in the same way when each type θ is a closed and convex subset of Γ, distinct types are disjoint, a type θ sender has a prior P^θ_δ over $\delta \in \theta$, and, if γ is not an extreme point of Γ, $P^\theta_\delta(\partial\theta \cap \Gamma^\circ) > 0$.

B.2 Overlapping Interval Types

Section 5 studies an accurate sender who coexists with a simple sender whose model fixes curvature at a single value. I now study general overlapping interval types. Previously, I assumed that senders differed in their ability to model the quadratic component of the receiver's models, that different senders could model this component in distinct ways, and thus that the receiver could fully back out the sender's type from the model she received. However, I now relax Assumption 2 and allow the sender's types to have overlapping supports. That is, I consider the case where there exist $\theta, \theta' \in \Theta$ such that $\theta' \cap \theta \neq \varnothing$.

Let $\delta \in \theta \cap \theta'$. Conditional on seeing a model of the form $y = \beta_0 + \beta_1 x - \delta x^2$, the receiver does not know whether this model came from a type θ or type θ' sender. In particular, I can characterize when uncertainty about the source of the signal can actually work to a sender's benefit, and when it hurts the sender.

Proposition 9. *Consider two senders θ, θ' such that $\theta \cap \theta' \neq \varnothing$. Suppose R receives information from either sender with positive probability. Then, a type θ sender's utility either strictly increases or decreases only if $\partial\theta' \cap \theta \neq \varnothing$ or $\partial\theta \cap \theta' \neq \varnothing$.*

There are many permutations to consider, and the proof of the proposition walks exhaustively through all the different cases, but we highlight three here. Denote $\theta = [\underline{\delta}, \overline{\delta}]$ and $\theta' = [\underline{\delta}', \overline{\delta}']$. Given $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$, a type θ sender's utility strictly *increases* from the existence of type θ' in the following cases.

1. $\delta \in \theta^\circ$, $\delta = \overline{\delta}'$, and θ' samples as a generalist.
2. $\delta \in \theta^\circ$, $\delta = \underline{\delta}'$, and θ' samples as a specialist.
3. $\delta = \overline{\delta}$, $\theta' = \{\delta\}$, type θ samples as a specialist, and θ' samples as a generalist.

In the first case, if $\delta \in \theta^\circ$, then f_S is the true model if it came from type θ. However, because $\delta = \overline{\delta}'$, if the model came from θ', the true model may also have $\gamma \geq \delta$. Accordingly, the type θ' sender samples as a generalist, meaning that R entertains models above f_S. Thus, the receiver entertains both the possibility that f_S is the true model, but also models with $\gamma \geq \overline{\delta}'$. This benefits the type θ sender, who knows that he has the true model.

The same logic applies to the second case, where $\delta = \underline{\delta}'$ and type θ' samples as a specialist. Type θ provides the true model, but the receiver entertains shallower models that lie above f_S thanks to the type θ' sender's local sampling.

The third case considers a situation where a type θ sender is restricted in how he can sample. In particular, it considers a case where $\delta = \overline{\delta}$ but the type θ sender is forced to sample as a specialist. This may occur because $\underline{\delta}$ is more likely for this sender type, and he thus ex-ante chooses specialist sampling, even though he (ex-post) would be better off sampling as a generalist. However, if $\theta' = \{\delta\}$ and that sender is sampling as a generalist, the type θ sender can be made better off, because the receiver now entertains some models above f_S rather than only ones below.

A corollary of this result is that interactions are zero-sum: if a type θ sender is made better off by a type θ', the type θ' sender is made worse off.

Corollary 2. *Conditional on providing a model $f_S(x)$, suppose a type θ sender's expected utility increases from a type θ' sender's existence. Then, the type θ' sender's expected utility decreases from the existence of a type θ sender.*

Proof of Proposition 9. We show the following cases exhaustively. Given $f_S(x)$, a type θ sender's utility strictly increases (decreases) from the existence of type θ' if one of the following holds.

1. $\delta \in \theta^\circ \neq \varnothing$, $\delta = \overline{\delta}'$, and θ' distantly (locally) samples.
2. $\delta \in \theta^\circ \neq \varnothing$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.
3. $\delta = \overline{\delta}$, $\theta^\circ = \varnothing$, type θ locally samples, and θ' distantly (locally) samples.

4. $\delta = \underline{\delta}$, $\theta^\circ = \varnothing$, type θ distantly samples, and θ' locally (distantly) samples.

5. $\theta^\circ = \varnothing$, $\theta'^\circ \neq \varnothing$, $\delta = \overline{\delta}'$, and θ' distantly (locally) samples.

6. $\theta^\circ = \varnothing$, $\theta'^\circ \neq \varnothing$, $\delta = \underline{\delta}'$, and θ' locally (distantly) samples.

First, we show how the receiver's expected value from entertaining models that have steeper or shallower tradeoffs than the sender's affects her maximal expected utility. Suppose $\gamma > \delta$. Given a sampling strategy σ, we have:

$$\max \beta_0 + \beta_1 x - \delta(x - x_\theta)^2 = \beta_0 + \beta_1 x_\theta + \frac{\beta_1^2}{4\delta}$$
$$\max \beta_0 + \beta_1 x - \gamma(x - x_\theta)^2 + (\gamma - \delta)\sigma^2 = \beta_0 + \beta_1 x_\theta + \frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2.$$

The latter is greater than the former when

$$\frac{\beta_1^2}{4\gamma} + (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4\delta} \iff (\gamma - \delta)\sigma^2 \geq \frac{\beta_1^2}{4}\left(\frac{1}{\delta} - \frac{1}{\gamma}\right) \iff \sigma^2 \geq \frac{\beta_1^2}{\gamma\delta}.$$

Now, suppose $\delta \in \theta_i^\circ, \theta_j^\circ$. Then, the receiver entertains just the sender's model, which she believes is the sender's (only) model. Thus, we consider situations where the boundaries of the sets intersect.

Next, suppose $\underline{\delta}_i \in \theta_j^\circ$. Conditional on seeing $\underline{\delta}_i$, the receiver entertains models with $\gamma \leq \underline{\delta}_i$. By Theorem 1, the sender would prefer to locally sample in these cases. However, in his initial choice of σ, the sender is forced to consider both the probability of $\overline{\delta}_i$ and $\underline{\delta}_i$. If $\sigma_i = \overline{\sigma}$, the sender is being forced to distantly sample, which delivers a gain most of the time but a loss in the case where $\underline{\delta}_i$ materializes. However, if $\underline{\delta}_i \in \theta_j^\circ$, then with some probability, the receiver believes the $\underline{\delta}_i$ model was the true model and came from the type j sender. This means the receiver places a higher weight on the sender's model itself, and less on models with lower γ values (which, under $\sigma_i = \overline{\sigma}$, deliver lower expected utility).

A similar argument shows that the sender *loses out* when $\delta = \underline{\delta}_i$ and $\sigma_i = \underline{\sigma}$. In this case, the sender is (optimally) gaining from his own local sampling, meaning the remaining models the receiver entertains deliver higher expected utility. However, the receiver thinks with positive probability that the sender's model is in fact the *true* model, causing her to relatively downweight these alternative models and focus more weight on the sender's model, which delivers lower expected utility. Yet another similar argument shows that the sender gains (loses) when $\overline{\delta}_i \in \theta_j^\circ$ and $\sigma_i = \underline{\sigma}$ ($\sigma_i = \overline{\sigma}$).

Next, suppose $\delta \in \theta_i^\circ$ and that $\delta = \overline{\delta}_j$. If the type j sender is distantly sampling ($\sigma_j = \overline{\sigma}$), the receiver entertains not only type i's model but also models with higher γ (since she believes the model could have also come from θ_j). The set of models R entertains is thus more optimistic than if she were to just entertain the type i sender's model, meaning the type i sender benefits from the type j's distant sampling. Local sampling, on the other hand, would hurt him. A similar

argument takes care of the case where $\delta = \underline{\delta}_j$, where the type θ_i sender is hurt by distant sampling and benefits from local sampling.

□

B.3 Designing Interval Type Spaces

In the design problem of Section 5, suppose that a can choose any type space $\Theta \subseteq \mathcal{P}([0, \overline{\gamma}])$ such that for all $\theta \in \Theta$, θ is a closed interval.

Proposition 10. *Suppose P^a_δ is continuous. Then, the optimal type space contains no non-degenerate intervals, and is the collection of singletons on $[0, \overline{\gamma}] = \Gamma$: $\Theta = \big\{\{\delta\} : \delta \in \Gamma\big\}$.*

Naturally, a could also introduce a sender of type $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$ with $\theta^\circ \neq \varnothing$. This sender would exaggerate some tradeoffs and neglect others, but also sometimes provide the correct model. In this case, the assumption that P^a_δ has a continuous density is crucial. The only time a could gain from this sender's existence is if that sender provided a model with either $\delta = \overline{\delta}$ or $\underline{\delta}$. However, a would not gain at all if $\delta \in (\underline{\delta}, \overline{\delta})$, since this type θ would always be providing the *true model*. This erases any potential gains from model simplicity and strategic sampling for a. Since $\overline{\delta}$ and $\underline{\delta}$ are single points, each point would have measure 0 in a's prior, while $(\underline{\delta}, \overline{\delta})$ has measure greater than 0. Thus, a would prefer to have many different senders model tradeoffs arbitrarily, rather than having other senders sometimes provide the true model and sometimes not.

Proof. Consider a sender model $f_S = \beta_0 + \beta_1 x - \delta x^2$. First, we show that Θ contains no non-degenerate intervals. Suppose by contradiction that there existed an interval of type $\theta = [\underline{\delta}_\theta, \overline{\delta}_\theta]$ with $\theta \subseteq \Gamma^\circ$. Suppose $\theta^\circ \neq \varnothing$ and consider $\delta \in \theta^\circ$. Conditional on seeing this model, the receiver believes the model $f_S(x) = \beta_0 + \beta_1 x - \delta x^2$ is the true model.

Now, for each $\delta \in \theta^\circ$, as before, a can create a type $\{\delta\}$. This type strategically samples by setting σ such that $\max_x f_S(x) \leq \max_x f_S(x) + (\mathbb{E}[\gamma|\delta])(\sigma^2 - (x - x_\theta)^2)$. Thus, conditional on seeing a model of the form $\beta_0 + \beta_1 x - \delta x^2$, R believes that $f_S(x)$ is either the true model or the model from the $\{\delta\}$ type. Let $q(\delta|\beta_0, \beta_1, T)$ be the conditional probability of the type $\{\delta\}$ sender relative to the original type θ sender. Let $\overline{\gamma}_{\{\delta\}}$ be the receiver's expectation of γ if $\delta \geq \gamma$ given f_S and $\overline{\gamma}_{\{\delta\}}$ likewise if $\delta < \gamma$. Then, this generates a difference on the order of

$$\int_{\underline{\delta}_\theta}^{\overline{\delta}_\theta} \max_x f_S(x) + q(\delta|\beta_0, \beta_1, T)\big(\overline{\gamma}_{\{\delta\}} - \delta\big)(\overline{\sigma}^2 - (x - x_\theta)^2) - \max_x f_S(x) dP^a_\delta(\delta \leq \mathbb{E}^R_\gamma[\gamma] | \delta \in \theta^\circ)$$
$$+ \int_{\underline{\delta}_\theta}^{\overline{\delta}_\theta} \max_x f_S(x) + q(\delta|\beta_0, \beta_1, T)\big(\underline{\gamma}_{\{\delta\}} - \delta\big)(\underline{\sigma}^2 - (x - x_\theta)^2) - \max_x f_S(x) dP^a_\delta(\delta > \mathbb{E}^R_\gamma[\gamma] | \delta \in \theta^\circ)$$

The first line considers any arbitrary senders who position as generalists. The second line considers any arbitrary senders who position as specialists. In both cases, the gain is positive and is increasing in $q(\delta|\beta_0, \beta_1, T)$; i.e., the gain is higher the more likely R is to think that $f_S(x)$ came from type $\{\delta\}$.

We are left to consider the boundary types of θ: $\overline{\delta}_\theta$ and $\underline{\delta}_\theta$. Since θ either samples as a generalist or specialist, by Proposition 9 one of $\overline{\delta}_\theta$ or $\underline{\delta}_\theta$ gains and one loses from the existence of an arbitrary type. However, because P^a_δ is continuous, these singletons have measure 0 with respect to the prior. Thus, replacing each with an arbitrary type has no effect on expected utility.

Hence, every type space Θ with non-degenerate intervals can be improved upon by replacing all the senders in that interval with singletons. The maximal set of singletons covers all of Γ, meaning $\Theta = \{\{\delta\} : \delta \in \Gamma\}$ maximizes the objective. □

B.4 Value of Information with Interval Types

With interval types, the receiver learns not only the slope $f'_R(x_\theta)$ but also whether γ lies below, within, or above the sender's interval. Let $\delta_\theta(\gamma) := \min\{\max\{\gamma, \underline{\delta}_\theta\}, \overline{\delta}_\theta\}$. Propositions 3 and 4 then hold with $\mathbb{E}^R_\gamma[\gamma|f'_R(x_\delta)]$ replaced by $\mathbb{E}^R_\gamma[\gamma|f'_R(x_\theta), \delta_\theta(\gamma)]$; the proofs are identical, conditioning throughout on $\delta_\theta(\gamma)$. In particular, if the receiver perfectly learns γ (i.e. $\delta_\theta(\gamma) = \gamma$), $V(\theta)$ collapses to the difference between $\mathbb{E}\left[\frac{\alpha_1^2}{\gamma}\right]$ and $\frac{\mathbb{E}[\alpha_1]^2}{\mathbb{E}[\gamma]}$.

B.5 Non-Quadratic Tradeoffs

While the main model utilizes quadratic utility to express tradeoffs, in this subsection, I probe the robustness of many of the main results to *non-quadratic* utility. In particular, I show an analogue of Theorem 1: if $\overline{\sigma}$ is sufficiently large and S estimates a naive model, sampling data as a generalist dominates sampling as a specialist. If S estimates a cautious model, being a specialist dominates being a generalist.

General Loss Function The following two assumptions consider two generalizations of functional forms for the receiver.

Assumption 4. *The receiver's models and a type θ sender's models are given by:*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma \ell(x - x_\theta) \tag{8}$$

$$f_S(x) = \beta_0 + \beta_1 x - \delta \ell(x - x_\theta) \qquad \theta \in \Theta \tag{9}$$

where $\ell : \mathbb{R} \to \mathbb{R}^+$ is strictly convex and differentiable. Moreover, $\ell(0) = 0$, $\ell'(0) = 0$, and ℓ is symmetric around $x = 0$.

One way to interpret this functional form is to think of a sender's or receiver's model as a *Taylor expansion* around a point x_θ. This Taylor expansion includes a constant component, a linear term, and a concave $\mathcal{O}(2)$ term centered at x_θ. While the convex loss $\ell(x - x_\theta)$ is symmetric around x_θ, the assumption does allow for f_R or f_S themselves to be asymmetric. This assumption subsumes the case where $\ell(x) = x^2$.

The case where this convex loss itself may be locally asymmetric is also of interest. Thus, I also consider the following alternative assumption.

Assumption 5. *The receiver's models and a type θ sender's models are given by:*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma m(x) \tag{10}$$

$$f_S(x) = \beta_0 + \beta_1 x - \delta m(x) \qquad \theta \in \Theta \tag{11}$$

where $m : \mathbb{R} \to \mathbb{R}^+$ is strictly convex and differentiable, and $m'(x)$ is onto $\mathbb{R}$. Moreover, there exists x such that $m(x) = m'(x) = 0$.

This formulation of models allows for a convex loss function m to take many different shapes. With these two assumptions in hand, I prove a more general version of Theorem 1.

Theorem 4. *Suppose Assumption 4 holds. If S estimates naive models, he positions himself as a generalist ($\sigma_\theta = \overline{\sigma}$) and sets $x_\theta = \max\{0, \min\left\{\frac{\mathbb{E}_1^\theta[\beta_1]}{2\mathbb{E}_\delta^\theta[\delta]}, \overline{x}\right\}\}$. If S estimates cautious models, he positions himself as a specialist ($\sigma_\theta = \underline{\sigma}$) and sets $x_\theta = 0$ or $\overline{x}$.*
Alternatively, suppose Assumption 5 holds, suppose $\overline{\sigma}$ is sufficiently large, and fix x_θ. If S estimates naive models, he strictly prefers *$\sigma_\theta = \overline{\sigma}$ to $\underline{\sigma}$. If S estimates cautious models, he* strictly prefers *$\sigma_\theta = \underline{\sigma}$ to $\overline{\sigma}$.*

First, we note that the first part of Theorem 4 is identical to Theorem 1: a sender who estimates naive models positions himself as a generalist, and a sender who estimates cautious models positions himself as a specialist. The key reason is that the set of plausible f_Rs must obey the following moment conditions.

$$\mathbb{E}_{\mu_\theta}\left[f_R(x) - \beta_0 - \beta_1 x + \delta \ell(x - x_\theta)\right] = 0 \tag{12}$$

$$\mathbb{E}_{\mu_\theta}\left[x\big(f_R(x) - \beta_0 - \beta_1 x + \delta \ell(x - x_\theta)\big)\right] = 0 \tag{13}$$

The proof shows that (13) can be boiled down to the condition that the derivative of all plausible f_Rs must be the same across the data sample μ_θ. Since μ_θ and $\ell(x - x_\theta)$ are both symmetric around x_θ, we have that for μ'_θ with $x_\theta = x'_\theta$ but $\sigma'_\theta \neq \sigma_\theta$

$$\mathbb{E}_{\mu_\theta}[f'_R(x)] = \alpha_1 - \gamma \mathbb{E}_{\mu_\theta}[\ell'(x - x_\theta)] = \alpha_1 - \gamma \mathbb{E}_{\mu'_\theta}[\ell'(x - x_\theta)] \qquad \forall \sigma, \sigma' \in [\underline{\sigma}, \overline{\sigma}]$$

Thus, equation (13) is invariant to σ. This means that, as in the quadratic case, as σ changes, the set of models that R considers as a function of σ shifts up or down, allowing us to back out the same logic as in the original case.

However, if models are not characterized by a symmetric loss $\ell(x - x_\theta)$ around x_θ, it is more difficult to obtain a result. Theorem 4 simply states that, fixing x_θ, S prefers being a generalist to being a specialist if he estimates naive models (and the opposite holds if he estimates cautious models). The reason is that, if the convex portion of a receiver's models is *not* symmetric around x_θ, the analogue of (13) no longer holds. I.e., the expression

$$\mathbb{E}_{\mu_\theta}[f'_R(x)] = \alpha_1 - \gamma \mathbb{E}_{\mu_\theta}[m(x)]$$

generally changes as the variance of a distribution μ_θ increases, fixing its mean. As a result, the proof allows α_1 to change to "absorb" any changes to $\mathbb{E}_{\mu_\theta}[m(x)]$ with respect to σ. For each $f_R(x)$ with a given γ value, changing σ thus not only changes α_0 (as in the benchmark and symmetric loss models) but also α_1. It is here where we utilize the condition in Assumption 5 that assigns the same prior mass to different α_1 values in the receiver's prior, which ensures that these models receive the same posterior weight.

Consequently, the value of the receiver's objective at her optimal choice x^* does not have an intuitive expression, and depends on nonintuitive features of the third derivative of $m(\cdot)$. However, the proof shows that if the sender estimates naive models, the value of each $f_R(x)$ the receiver entertains at x_θ is *increasing* in σ. Hence, the value of each f_R at x_θ for sufficiently high $\overline{\sigma}$ must necessarily be higher than the receiver's optimum x^* at $\underline{\sigma}$. The intuition for the case where the sender estimates cautious models, as before, is flipped.

Proof of Theorem 4 We prove this result via the following lemmata. The first part of the theorem (assuming Assumption 4) follows immediately from the first lemma after applying the same reasoning as in Theorem 1.

Lemma 1. *Suppose Assumption 4 holds and fix f_S. Then, for all $f_R \in \phi(f_S; \theta, \mu_\theta)$, there exists $L(\sigma)$ strictly increasing in σ with $L(0) = 0$ such that:*

$$f_R(x) = f_S(x) + (\gamma - \delta)\big[L(\sigma) - \ell(x - x_\theta)\big).$$

Proof. Suppose $f_R \in \phi(f_S; \theta, \mu_\theta)$ for $f_S = \beta_0 + \beta_1 x - \delta\ell(x - x_\theta)$. This means that we have

$$f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1 x - \gamma\ell(x - x_\theta).$$

Applying Stein's Lemma as above, the moment condition with respect to $\tilde{\alpha}_1$ can be written as

$$\begin{aligned}\mathbb{E}_{\mu_\theta}\big[x\big(f_R(x) - f_S(x)\big)\big] &= \mathrm{Cov}_{\mu_\theta}(x, f_R(x) - f_S(x)) \\ &= \sigma^2\mathbb{E}_{\mu_\theta}[f_R'(x) - f_S'(x)] \\ &= \sigma^2\mathbb{E}_{\mu_\theta}[\tilde{\alpha}_1 - \gamma\ell'(x - x_\theta) - \beta_1 + \delta\ell'(x - x_\theta)] = 0.\end{aligned}$$

Because $\ell(x - x_\theta)$ is symmetric around x_θ, the terms with ℓ' are equal to 0. Thus, we have $\tilde{\alpha}_1 = \beta_1$ for all σ and are left to solve for $\tilde{\alpha}_0$. But for each γ value, as above, we have

$$\begin{aligned}\mathbb{E}_{\mu_\theta}[f_R(x)] = \mathbb{E}_{\mu_\theta}[f_R(x)] &\implies \tilde{\alpha}_0 - \gamma\mathbb{E}_{\mu_\theta,\sigma}[\ell(x - x_\theta)] = \beta_0 - \delta\mathbb{E}_{\mu_\theta,\sigma}[\ell(x - x_\theta)] \\ &\implies \tilde{\alpha}_0 = \beta_0 + (\gamma - \delta)\mathbb{E}_{\mu_\theta,\sigma}[\ell(x - x_\theta)].\end{aligned}$$

Defining $L(\sigma) = \mathbb{E}_{\mu_\theta,\sigma}[\ell(x - x_\theta)]$, we note that because $\ell(x - x_\theta)$ is convex, $L(\sigma)$ is increasing in σ. As $\sigma \to 0$, $L(\sigma) \to 0$. This gives the result. □

Lemma 2. *Suppose Assumption 5 holds and fix f_S. Then, for each $f_R \in \phi(f_s; \theta, \mu_\theta)$, $f_R(x_\theta)$ is increasing (decreasing).*

Proof. As above, we write $f_R(x) = \tilde{\alpha}_0 + \tilde{\alpha}_1 x - \gamma m(x)$. Applying Stein's Lemma again to the slope moment condition, we have $\mathbb{E}_{\mu_\theta}[f_R'(x)] = \mathbb{E}_{\mu_\theta}[f_S'(x)]$. Using this condition, we implicitly define $\alpha_1(\sigma)$ as

$$\alpha_1(\sigma) = \beta_1 + (\gamma - \delta)\mathbb{E}_{\mu_\theta}[m'(x)]$$

and note that its sign with respect to σ is ambiguous. Similarly, we define $\alpha_0(\sigma)$ implicitly as

$$\alpha_0(\sigma) = \beta_0 + \big(\beta_1 - \alpha_1(\sigma)\big)x_\theta + (\gamma - \delta)\mathbb{E}_{\mu_\theta,\sigma}[m(x)]$$

and note in particular that, by definition, we have

$$\mathbb{E}_{\mu_\theta,\sigma}[f_S(x)] = \mathbb{E}_{\mu_\theta,\sigma}[f_R(x)] = \mathbb{E}_{\mu_\theta,\sigma}\big[\alpha_0(\sigma)x + \alpha_1(\sigma) - \gamma m(x)\big]$$

But then, by the envelope theorem, we have that

$$\begin{aligned}
\frac{\partial}{\partial\sigma} f_R(x_\theta|\sigma) &= \alpha_0'(\sigma) + \alpha_1'(\sigma)x_\theta - \gamma m(x_\theta) \\
&= -\alpha_1'(\sigma)x_\theta + (\gamma - \delta)\frac{\partial}{\partial\sigma}\mathbb{E}_{\mu_\theta,\sigma}[m(x)] + \alpha_1'(\sigma)x_\theta \\
&= (\gamma - \delta)\frac{\partial}{\partial\sigma}\mathbb{E}_{\mu_\theta,\sigma}[m(x)]
\end{aligned}$$

Since m is convex, this is weakly positive if and only if $\gamma \geq \delta$. Moreover, as $\sigma \to \infty$, $\mathbb{E}_{\mu_\theta,\sigma}[m(x)]$ is bounded away from 0, meaning that $f_R(x_\theta)$ approaches ∞ as $\sigma \to \infty$.

We note that because the sender behaves as if the receiver has a flat prior over α_0 and α_1, her posterior places the same mass on all models that are the same "up to a constant and linear term." This means that for $\gamma > \delta$, there exists σ such that $f_R(x_\theta|\sigma) > f_R(x^*|\underline{\sigma})$, where x^* maximizes $f_R(x|\underline{\sigma})$. Hence, S is better off being a generalist than a specialist when $\gamma > \delta$. The same logic shows that S prefers being a specialist when $\gamma < \delta$. Applying the same logic as in the proof of Theorem 1 gives the result. □